Button

DETECT | PICO BLOCKS | INPUT

What does it do?

A button lets someone tell your program to do something the moment they press.

Inside are two metal contacts. Pressing pushes them together so electricity can flow, and letting go springs them apart again.

Your projects can use it to start something, stop something, or count how many times it has been pressed.

Uses

Pedestrian crossing

DETECT Button and traffic phase → **DECIDE** Is it safe to cross? → **DO** Change the lights

Do not press this

DETECT Button is pressed → **DECIDE** Is the trap armed? → **DO** Play a silly sound

Cap
Press here

Legs
Four, joined in pairs

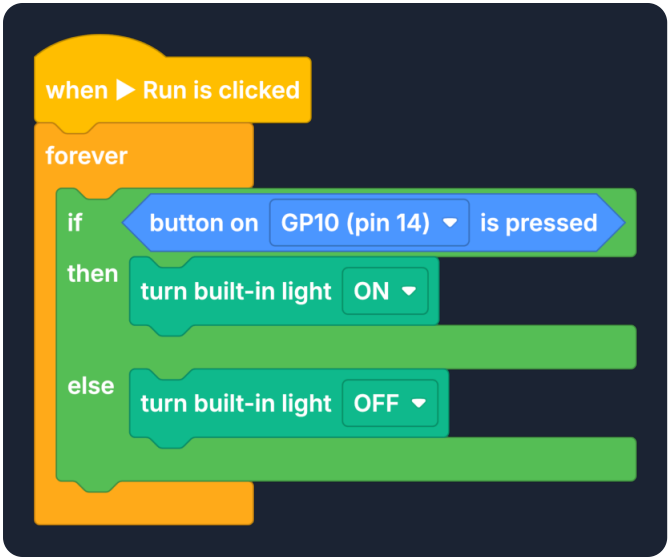
Gap
Straddle the channel

Raspberry Pi Foundation

Blocks

MicroLab

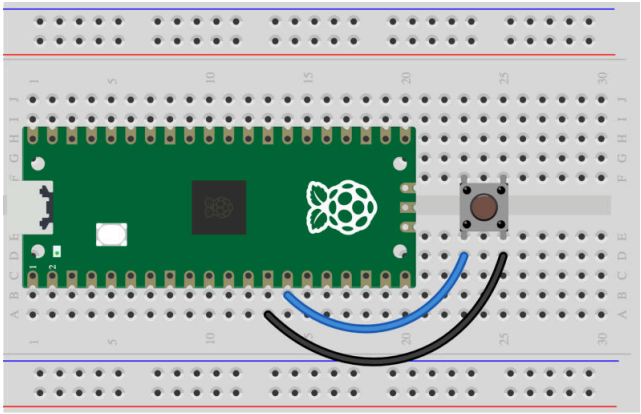
getmicrolab.com



The Pico's own light stays on while you hold the button down.

Wiring

PICO



GP10 (pin 14) → button → **GND** (pin 13)

Capacitive touch

DETECT | PICO BLOCKS | INPUT

What does it do?

A capacitive touch sensor notices when a finger comes close to its metal pad.

It detects the tiny change in electric charge caused by your body, so no moving parts are needed.

Use it anywhere a project needs a hidden, wipe-clean or feather-light button.

Uses

Touch lamp

DETECT: Finger touches the pad → DECIDE: Is the lamp already on? → DO: Toggle the light

Banana piano

DETECT: A banana is touched → DECIDE: Which note should play? → DO: Play the note

Touch pad
The large metal spiral

SIG
Digital output

VCC and GND
3.3 V power

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Blocks

MicroLab

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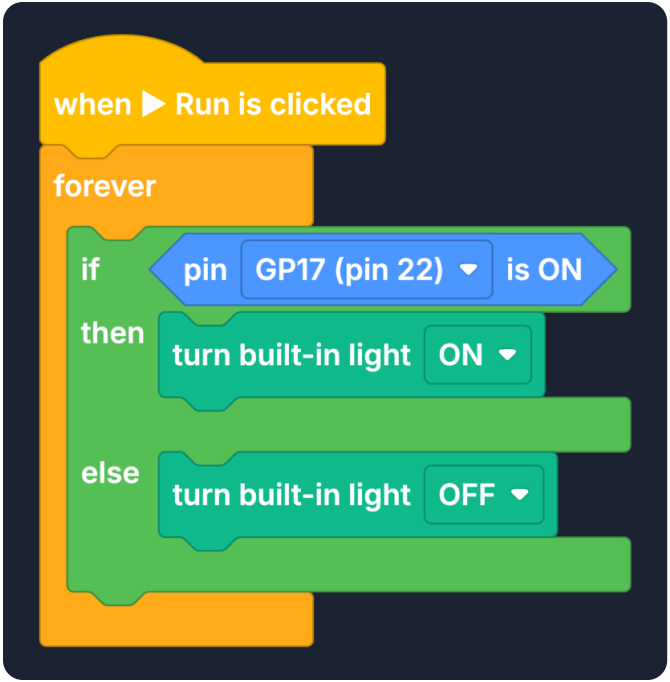
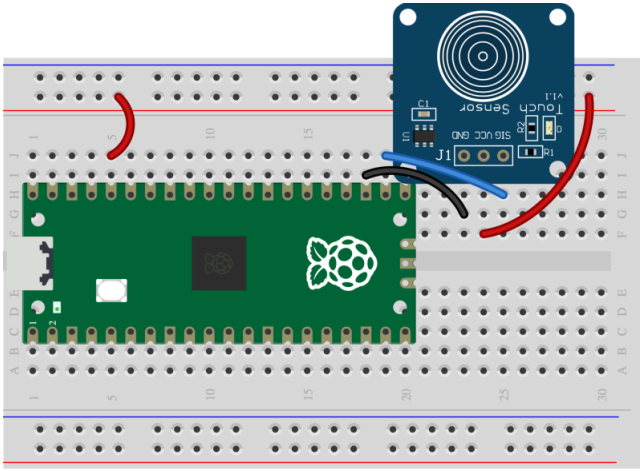


The Pico's own light comes on while you touch the pad.

Wiring

PICO



GND (pin 23) → GND, **3V3 OUT** (pin 36) → VCC, **GP17** (pin 22) → SIG

LDR

DETECT | PICO BLOCKS | LIGHT

What does it do?

An LDR is a sensor that measures how bright or dark it is.

LDR stands for light-dependent resistor. Light shining on it lets electricity through more easily, so your code can tell light from dark.

Your projects can use it to react when a lamp comes on, when night falls, or when a shadow passes over.

Uses

Street light

DETECT How dark it is → **DECIDE** Is it darker than dusk? → **DO** Switch the lamp on

Biscuit tin alarm

DETECT Light inside the tin → **DECIDE** Is the lid open? → **DO** Sound the alarm

Window Light lands here

Track Zigzag sensor

Legs Either way round

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Blocks

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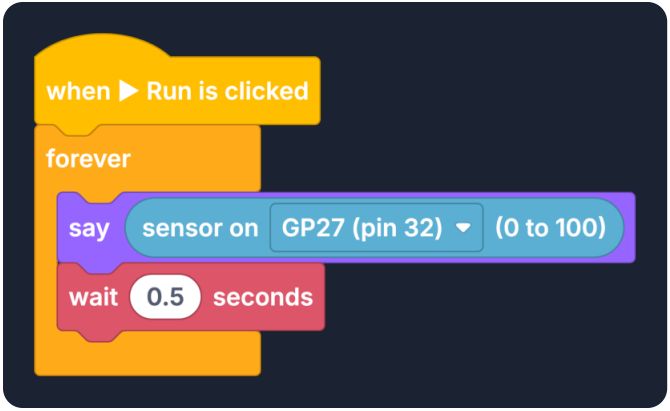
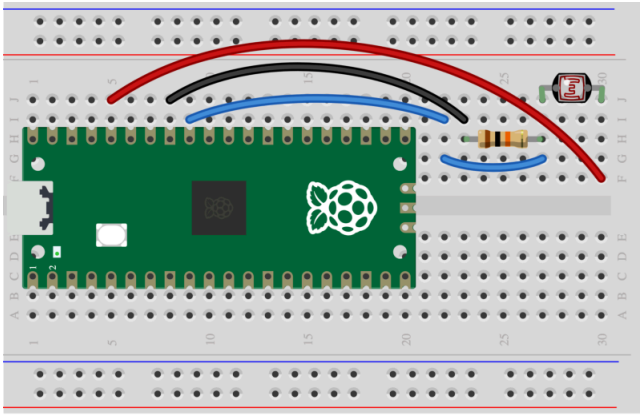


The number goes up in bright light and down in the dark, from 0 to 100. Watch it in the terminal.

Wiring

PICO



3V3 (pin 36) → LDR → **GP27** (pin 32) → 10 kΩ → **GND** (pin 33)

Line sensor

DETECT | PICO BLOCKS | INPUT

What does it do?

A line sensor tells light and dark surfaces apart underneath a moving project.

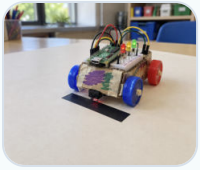
An infrared LED shines at the floor and a detector measures how much light bounces back.

Adjust the small potentiometer until its digital output changes reliably at the line edge.

Uses

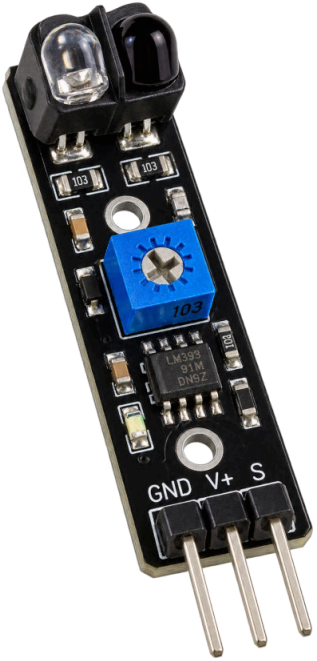
Warehouse guide-line robot

DETECT Read guide-line position → **DECIDE** Is the vehicle drifting? → **DO** Correct the steering



Cardboard lap-counting car

DETECT Cross dark finish strip → **DECIDE** Is this a new lap? → **DO** Increase the lap count



Infrared pair
Faces the floor

Adjuster
Sets the threshold

S
Digital signal

Raspberry Pi Foundation

Blocks

MicroLab

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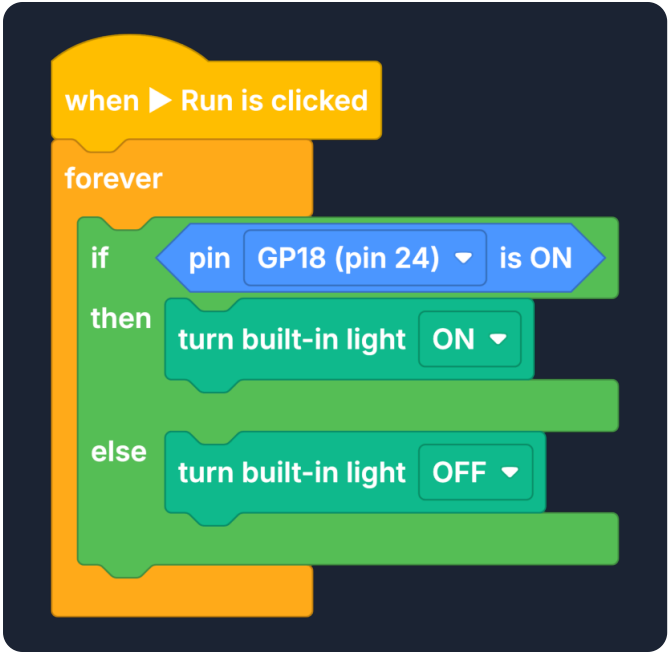
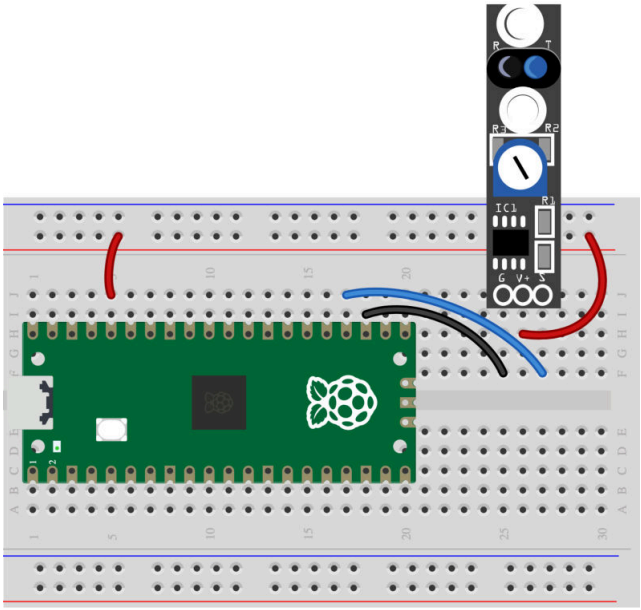


The Pico's light comes on over the dark line. Turn the blue adjuster until it changes cleanly at the line's edge.

Wiring

PICO



GND (pin 23) → G, **3V3 OUT** (pin 36) → V+, **GP18** (pin 24) → S

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Distance sensor

DETECT | PICO BLOCKS | DISTANCE

What does it do?

An HC-SR04P distance sensor measures how far away something is without touching it.

It works like a bat. One side sends out a click too high for you to hear, the other side listens for the echo, and the time it takes gives the distance.

Your projects can use it to react as something comes closer, or to measure a gap that keeps changing.

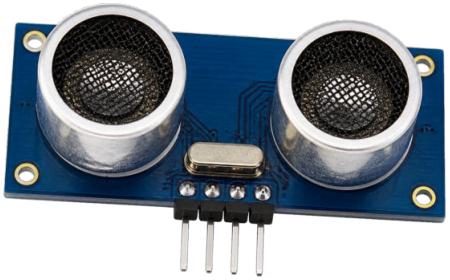
Uses

Parking sensor

DETECT: Gap to the wall → DECIDE: Is it closer than 30 cm? → DO: Beep faster

Air harp

DETECT: How far your hand is → DECIDE: Which note is that? → DO: Play the note



VCC	Trig	Echo	GND
Power, 3V3	Sends a click	Times the return	Ground

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Blocks

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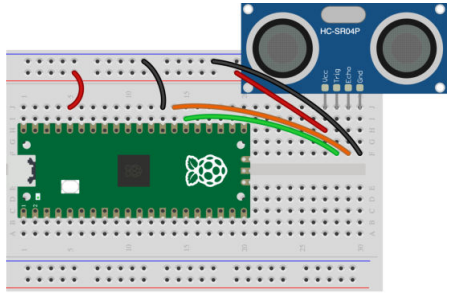
Shows centimetres. 400 means nothing is in range.

Wiring

PICO



HC-SR04P · Trig **GP20** (pin 26) · Echo **GP21** (pin 27) · **3V3 OUT** (pin 36) to top + rail · **GND** (pin 28) to top − rail

PIR

DETECT | PICO BLOCKS | MOTION

What does it do?

A PIR spots movement nearby, such as somebody walking past.

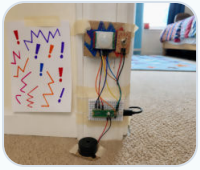
PIR stands for passive infrared. It watches the warmth that bodies give off and reports when that warmth moves, so it notices people and animals rather than furniture.

Your projects can use it to wake something up the moment somebody arrives, and go quiet again when they leave.

Uses

Burglar alarm

DETECT Movement in the room → **DECIDE** Should the room be empty? → **DO** Raise the alarm



Sibling alarm

DETECT Movement at the door → **DECIDE** Is privacy mode armed? → **DO** Sound the warning

Dome Wide-angle lens

VCC Power, 5V

OUT High on movement

GND Ground

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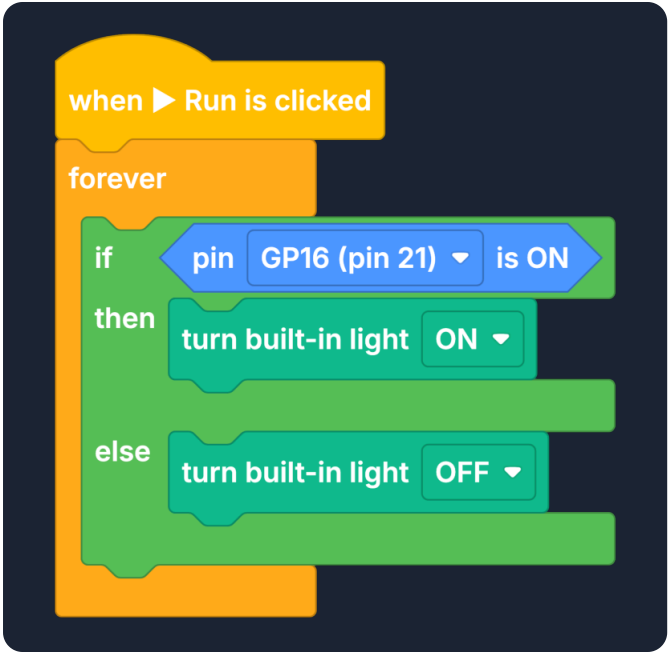
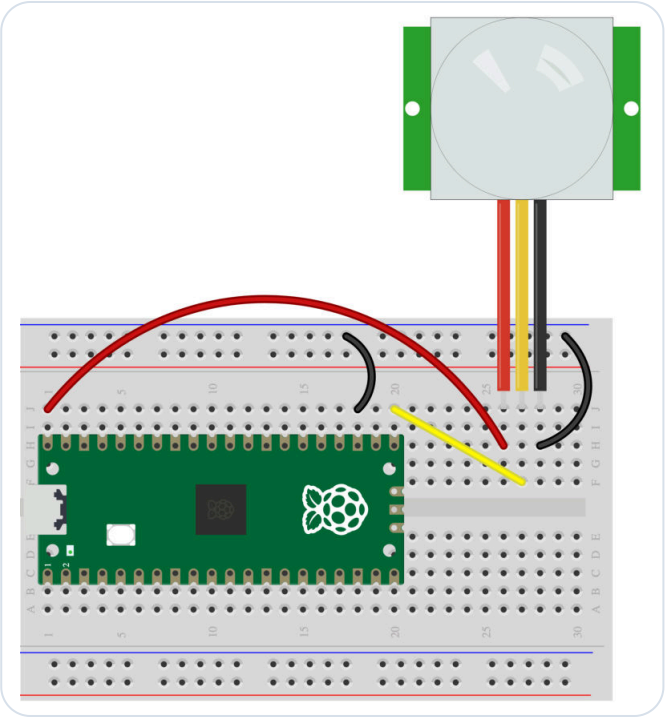


The Pico's light comes on when something warm moves nearby. After power-up, give the sensor a minute to settle.

Wiring

PICO



VBUS (pin 40) · OUT **GP16** (pin 21) · **GND** (pin 23)

AM312 PIR

DETECT | PICO BLOCKS | MOTION | MINI PIR

What does it do?

An **AM312** is a tiny PIR sensor that notices warm bodies moving nearby.

PIR means passive infrared. Its faceted lens divides the view into zones, so a person or animal moving between them makes the output switch on.

The complete AM312 module runs from 3V3 and uses very little power. Its sensitivity and roughly two-second output pulse are fixed.

Uses

Wardrobe light

DETECT Movement in the wardrobe → **DECIDE** Is someone reaching in? → **DO** Turn on the light

Dragon treasure alarm

DETECT Movement near the hoard → **DECIDE** Is someone sneaking closer? → **DO** Make the dragon roar

Lens	VCC	OUT	GND
Splits the view into zones	Power from 3V3	High on movement	Ground

Raspberry Pi Foundation

Blocks

MicroLab

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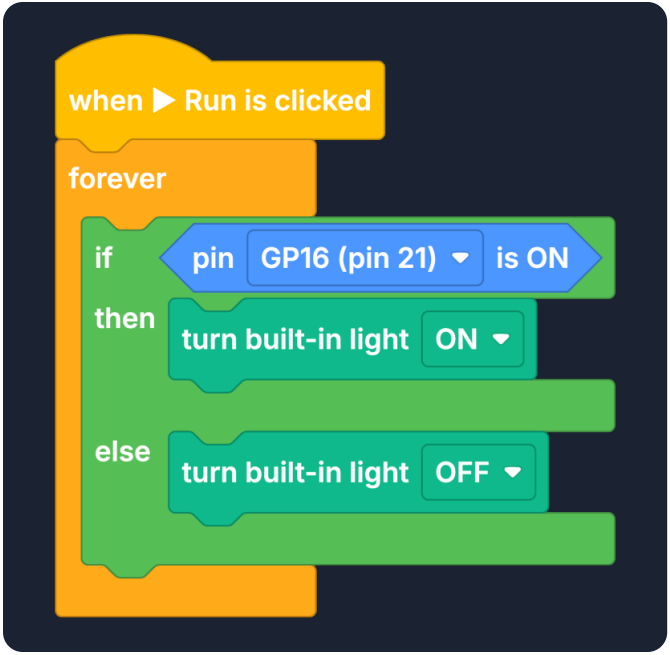
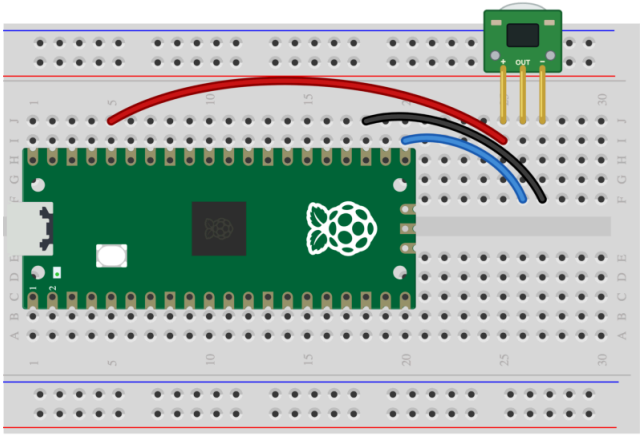


The Pico's light comes on when something warm moves nearby. After power-up, give the sensor a minute to settle.

Wiring

PICO



VCC → 3V3 OUT (pin 36); **OUT** → GP16 (pin 21); **GND** → GND (pin 23).

Potentiometer

DETECT | PICO BLOCKS | INPUT

What does it do?

A potentiometer lets your code read how far a knob has been turned.

This small version is called a trimpot. Turning the blue knob moves a contact called the wiper, changing the voltage your board reads.

Use it to choose a speed, a brightness or a sound. Turn gently and stop at the ends. The marking 103 means 10 kΩ.

Uses

Security-light adjustment

DETECT: Read sensitivity setting → DECIDE: Is movement strong enough? → DO: Switch the light on

Alien voice tuner

DETECT: Knob position → DECIDE: Should it rumble or squeak? → DO: Rumble or squeak

Knob	Outer pair	Wiper
Turn to adjust	Pins 1 and 3	Pin 2, offset

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Blocks

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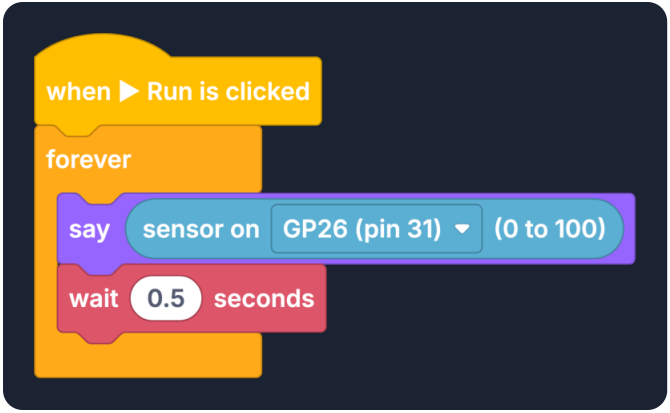
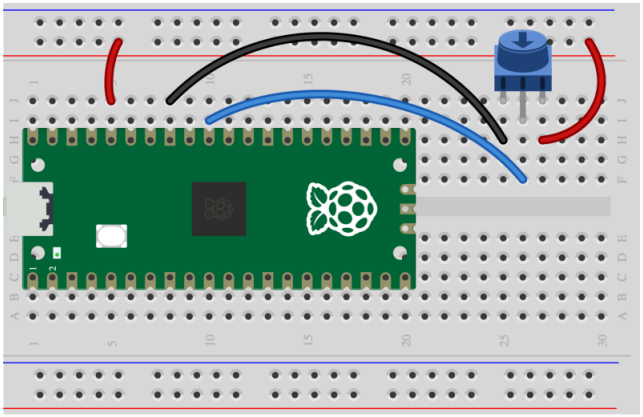


sensor on GP26 gives 0 to 100 as you turn the knob, and **say** shows the number in MicroLab's terminal. Power the trimpot from **3V3**.

Wiring

PICO



Trimpot pin **1** → **GND** (pin 33) · **2** (wiper) → **GP26** (pin 31) · **3** → **3V3 OUT** (pin 36). Each pin has its own breadboard row.

Temperature sensor

DETECT | PICO BLOCKS | INPUT | ANALOGUE

What does it do?

A TMP36 turns temperature into a changing voltage that a program can measure.

Its output rises by 10 millivolts for every degree Celsius, with 0.5 V representing 0 °C.

The Pico reads that voltage directly; the Raspberry Pi uses an MCP3008 analogue-to-digital converter.

Uses

Refrigerator cooling

DETECT Measure fridge temperature → **DECIDE** Is it too warm? → **DO** Increase the cooling

Soup drama meter

DETECT Measure bowl surface → **DECIDE** Which drama level fits? → **DO** Move the drama pointer

Flat face	+Vs	Vout	GND
Use it to identify the pins	Left pin	Centre pin	Right pin

Raspberry Pi Foundation

Blocks

MicroLab

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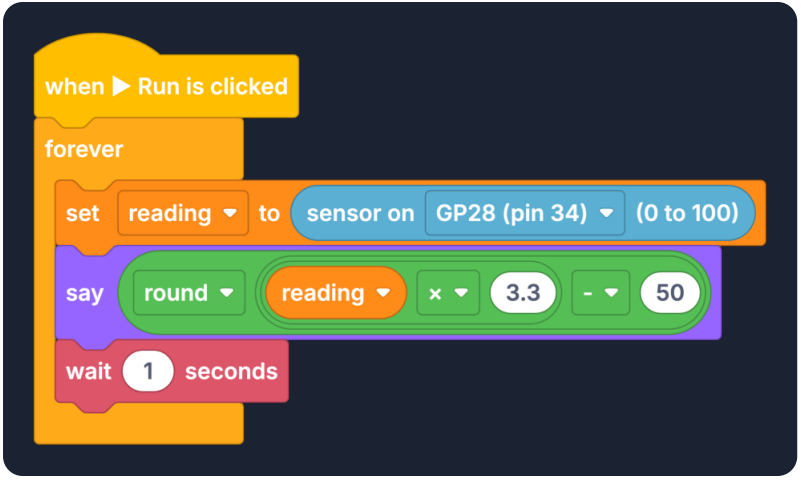
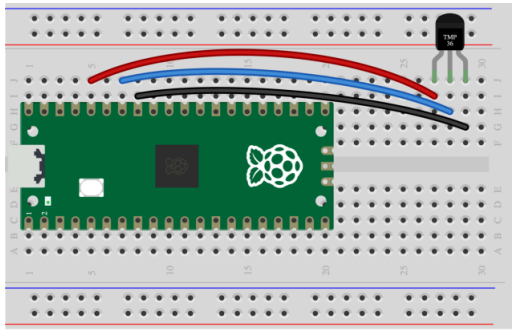


Shows °C in steps of about 3 °C. Make a variable called reading first.

Wiring

PICO



TMP36 **+Vs** → 3V3 OUT (pin 36), **Vout** → GP28 (pin 34), **GND** → GND (pin 33)

Tilt switch

DETECT | PICO BLOCKS | INPUT

What does it do?

A tilt switch changes state when it tips past a particular angle.

While it stands upright, a tiny metal ball rests on two contacts and completes the circuit. Tip it over and the ball rolls away, breaking the circuit.

It can detect that an object has been picked up, tipped over or opened.

Uses

Iron safety shut-off

DETECT Iron tips sideways → **DECIDE** Has the delay elapsed? → **DO** Switch the heater off

Grumpy pencil case

DETECT Case is tipped → **DECIDE** Has it stayed tipped? → **DO** Make a grumpy noise

Metal can
Contains the rolling ball

Legs
No polarity

Angle
Changes the contact

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Blocks

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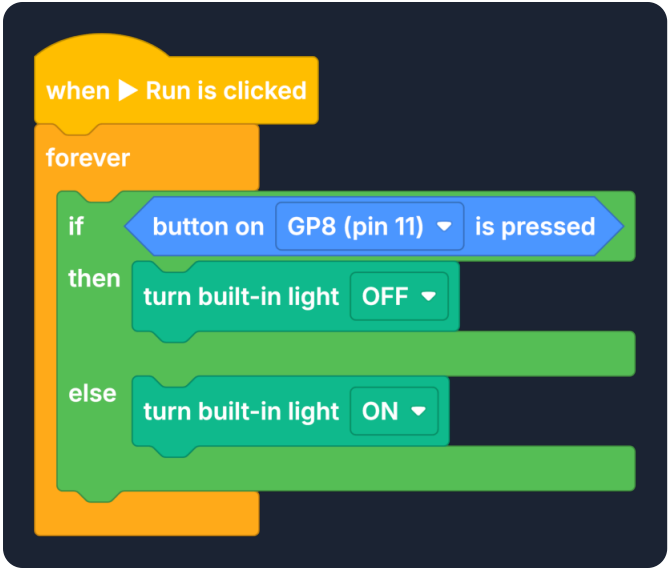
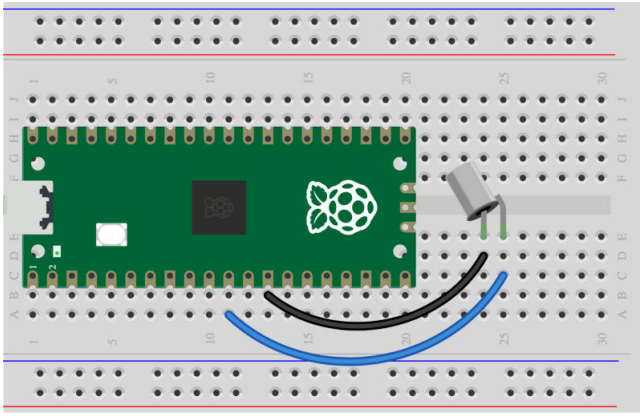


The Pico's own light comes on when you tip the switch over.

Wiring

PICO



GP8 (pin 11) → tilt switch → **GND** (pin 13)

Toggle switch

DETECT | PICO BLOCKS | INPUT

What does it do?

A toggle switch stays in the position you move it to, unlike a push button.

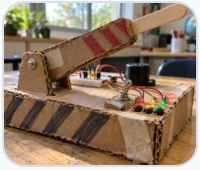
Its centre COM leg connects to one outside leg in each position.

It is useful when a project needs a lasting on/off choice or two modes.

Uses

Auto/manual selector

DETECT Read selector position → **DECIDE** Is automatic mode selected? → **DO** Run the selected mode



Serious business lever

DETECT Lever and check buttons → **DECIDE** Are all checks complete? → **DO** Start unnecessary sirens

Lever Stays where it is moved

COM Centre pin

L1 and L2 The two choices

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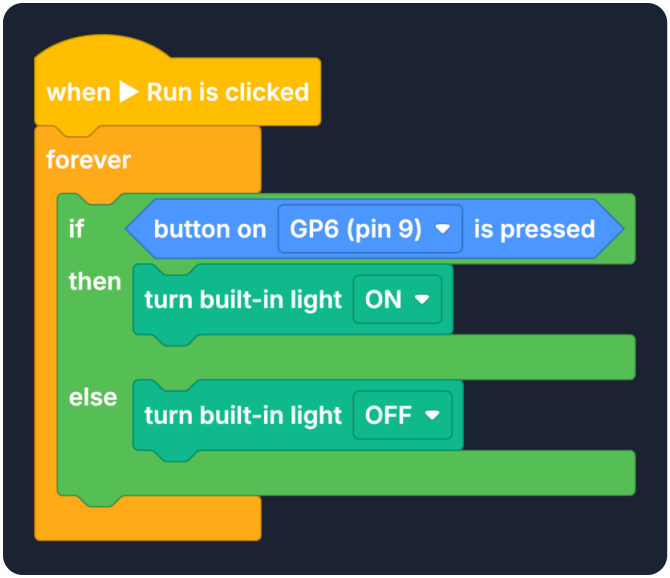
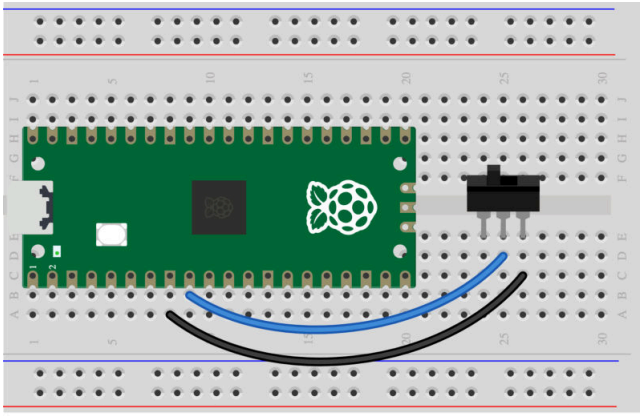


MicroLab calls a switch wired to GND a button: "pressed" means the switch is on. The Pico's light shows which way it is set.

Wiring

PICO



GP6 (pin 9) → COM; right-hand outside pin → **GND** (pin 8)

LED

DO | PICO BLOCKS | LIGHT

What does it do?

An LED is a tiny light that your code can switch on, switch off, or blink.

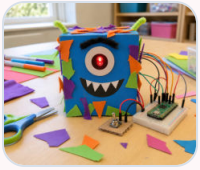
LED stands for light-emitting diode. It glows when electricity flows the right way through it, so the long leg and the short leg matter.

Your projects can use LEDs to show a warning, celebrate something, or tell people what is happening.

Uses

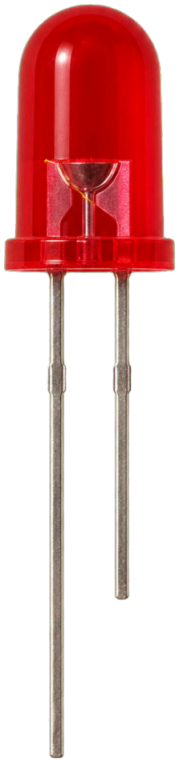
Car brake light

DETECT Brake control signal → **DECIDE** Is braking requested? → **DO** Light the red LEDs



One-eyed monster box

DETECT Box lid is opened → **DECIDE** Is guard mode armed? → **DO** Glow the monster eye



Anode
Long leg, to GPIO

Cathode
Short leg, to GND

Lens
Glows here

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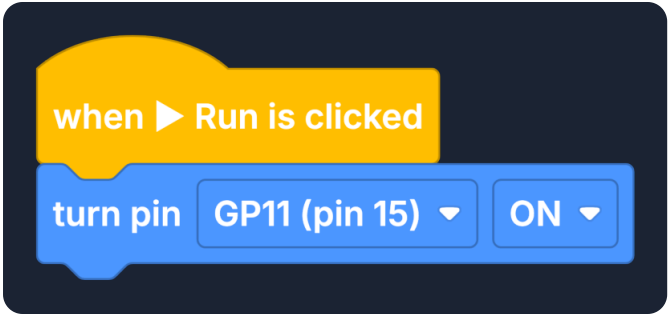
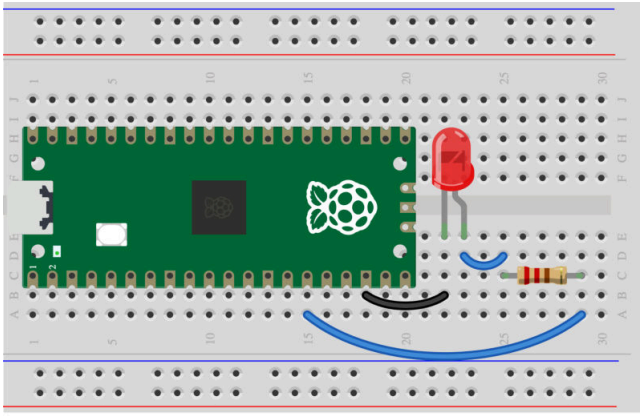


Change ON to OFF to switch the LED off again.

Wiring

PICO



GP11 (pin 15) → 220 Ω → LED → **GND** (pin 18)

RGB LED

DO | PICO BLOCKS | OUTPUT | LIGHT

What does it do?

An RGB LED puts red, green and blue lights inside one clear package.

Varying the brightness of the three colours mixes millions of possible colours.

A common-cathode LED has one shared negative leg and needs one resistor for each colour.

Uses

Wi-Fi router status

DETECT Read connection status → **DECIDE** Which status colour is needed? → **DO** Mix the LED colour

Mood crystal

DETECT Read the mood dial → **DECIDE** What is the crystal's mood? → **DO** Glow that colour

Clear lens
Contains three LEDs

Long leg
Common cathode

R, G and B legs
One 220 ohm resistor each

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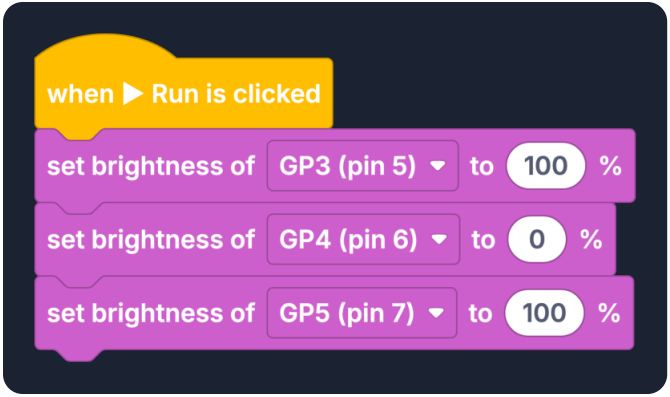
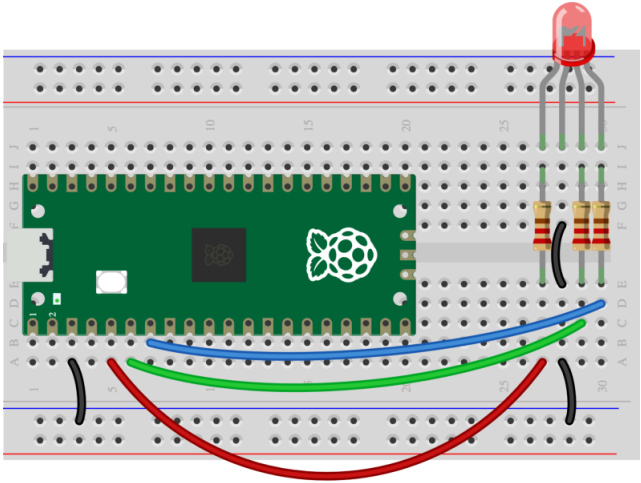


Brightness runs from 0 to 100 %: GP3 is red, GP4 green and GP5 blue. This mix makes magenta.

Wiring

PICO



R → **GP3** (pin 5), G → **GP4** (pin 6), B → **GP5** (pin 7); common → **GND** (pin 3)

NeoPixels

DO | PICO BLOCKS | OUTPUT | LIGHT

What does it do?

A NeoPixel is a full-colour LED with a tiny controller built into it.

One data wire carries colour instructions, and each pixel passes the remaining instructions onwards.

The 3.3 V-compatible JST breakout shown here can connect directly to a Pi or Pico.

Uses

Gaming PC lighting

DETECT Read lighting selection → **DECIDE** Which colour should each pixel be? → **DO** Light individual pixels

Disco shoelace meter

DETECT Read shoe movement → **DECIDE** Which light pattern should run? → **DO** Animate the pixels

Pixel RGB LED and controller

IN Data from the board

OUT Data to another pixel

VDD and GND 3.3 V power

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MicroLab

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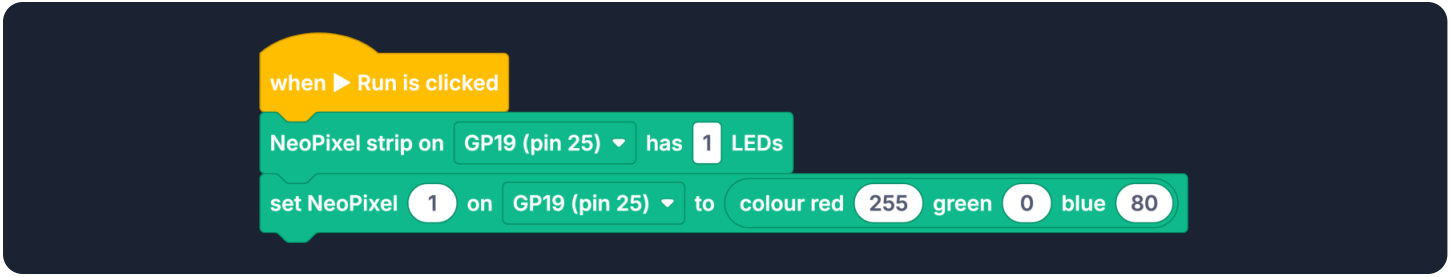
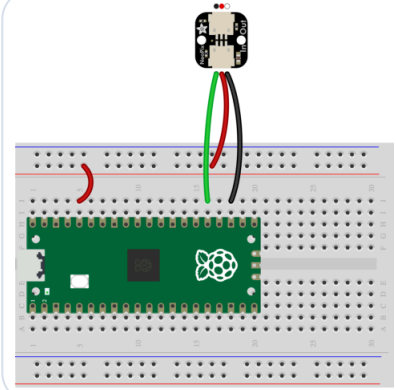


Change the red, green and blue numbers (0 to 255) to mix a new colour.

Wiring

PICO



GP19 (pin 25) → IN, **3V3 OUT** (pin 36) → top + rail → VDD, **GND** (pin 23) → GND

Active buzzer

DO | PICO BLOCKS | SOUND

What does it do?

An active buzzer makes one built-in tone. Switch it on and off for warning beeps or rhythms; its pitch stays the same.

Recognise it: check the packet for **active** or **built-in oscillator**. Shape, stickers and + marks do not prove the type.

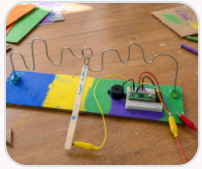
Use a labelled **3.3 V GPIO-compatible** buzzer.

Uses

Smoke alarm

One warning tone is enough; no changing notes needed.

DETECT Measure smoke level → **DECIDE** Is smoke above the limit? → **DO** Repeat one warning tone



Buzz wire game

Choose active: one fixed buzz clearly signals failure.

DETECT Loop touches wire → **DECIDE** Is this the third strike? → **DO** Buzz at one pitch

+ leg
Marked +, to GPIO

– leg
Other leg, to GND

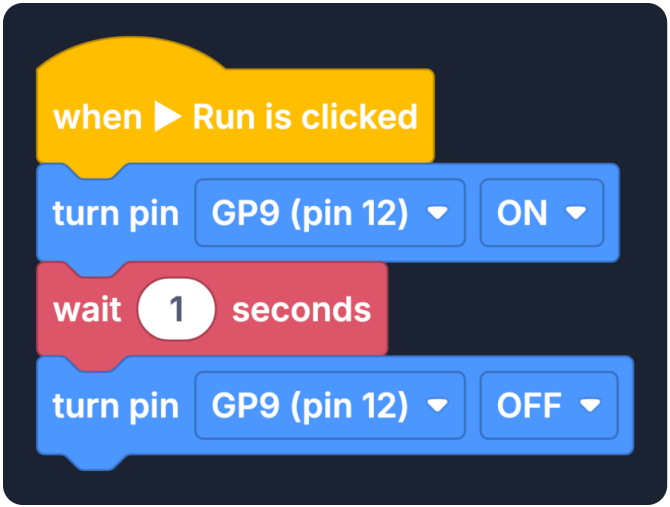
Vent
Sound comes out

Raspberry Pi Foundation

Blocks

MicroLab

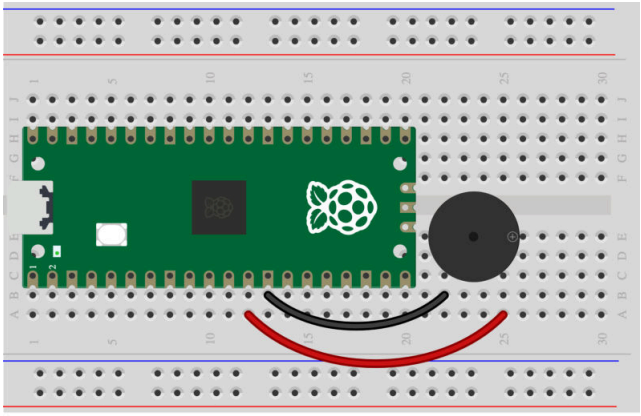
getmicrolab.com



It beeps for one second. Change the wait to make the beep longer.

Wiring

PICO



GP9 (pin 12) → **+** buzzer **−** → **GND** (pin 13)

Passive buzzer

DO | PICO BLOCKS | SOUND | PWM

What does it do?

A passive buzzer needs fast pulses (PWM) to make sound. Changing their frequency changes its pitch: low or high notes.

Recognise it: check for **passive**, **external drive** or **no oscillator** on the packet. Both types can look identical; + only marks polarity.

Use a small **3.3 V GPIO-compatible piezo** sounder.

Uses

Washing-machine chime

PWM plays tunes; active buzzers have only one pitch.

DETECT Read cycle status → **DECIDE** Has the wash finished? → **DO** Play a short tune

Robot voice box

Change pitch for happy chirps or grumpy growls.

DETECT Read mood button → **DECIDE** Is happy pressed? → **DO** Chirp high or low

+ leg
Marked +, to GPIO

– leg
Other leg, to GND

Vent
Sound comes out

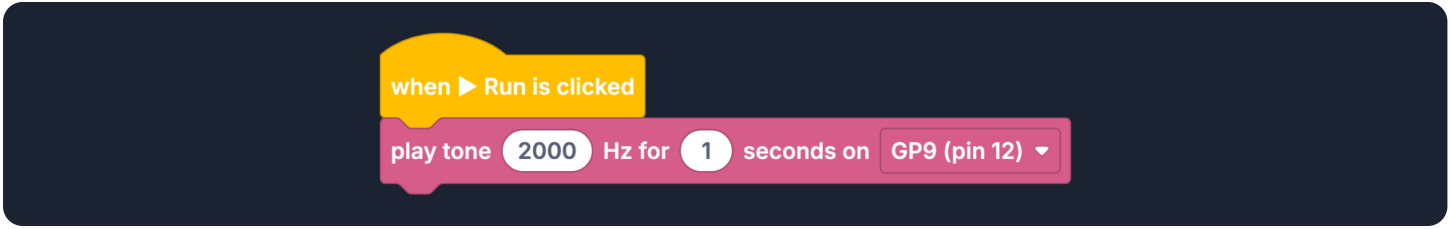
Raspberry Pi Foundation

Blocks

Wiring

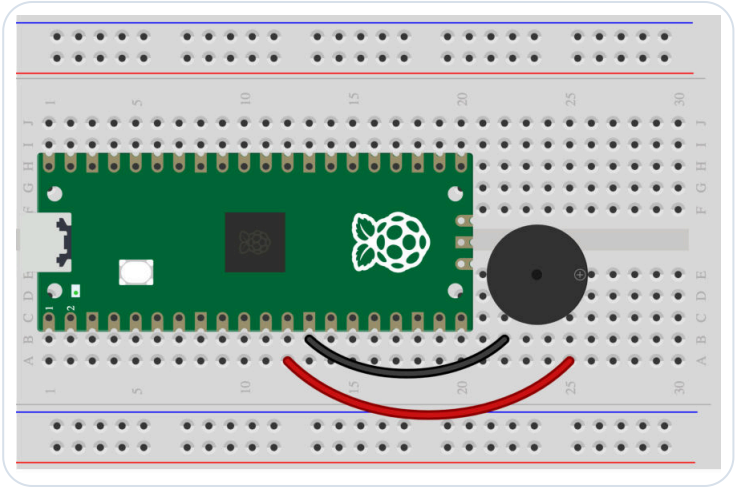
MicroLab

getmicrolab.com



Change 2000 Hz to play a higher or lower tone.

PICO



GP9 (pin 12) → **+** buzzer **−** → **GND** (pin 13)

Speaker

DO | PICO BLOCKS | OUTPUT | SOUND

What does it do?

A small passive speaker turns a changing electrical signal into sound.

Current through its coil moves a cone backwards and forwards, pushing the air.

The 220 ohm resistor shown here protects the GPIO; it also keeps this simple circuit fairly quiet.

Uses

Door chime

DETECT Button and chime setting → **DECIDE** Which chime should play? → **DO** Play the notes

Tiny fanfare machine

DETECT Count achievement-button presses → **DECIDE** Has the target been reached? → **DO** Play triumphant notes

Cone
Pushes the air

Coil and magnet
Create movement

Terminals
Connect through 220 ohms

Raspberry Pi Foundation

Blocks

MicroLab

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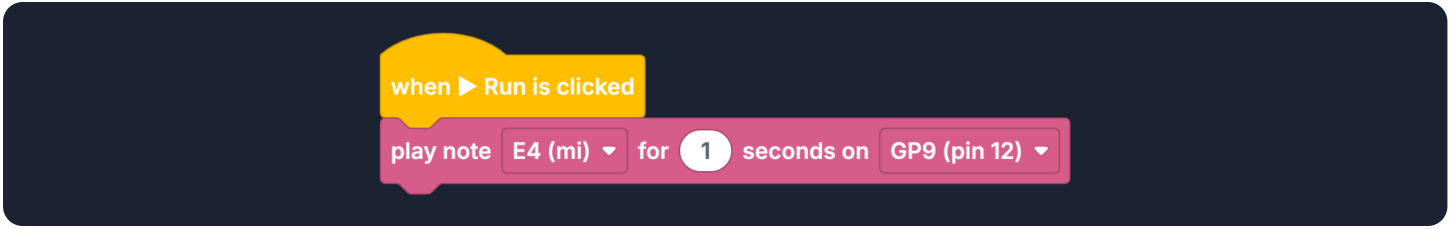
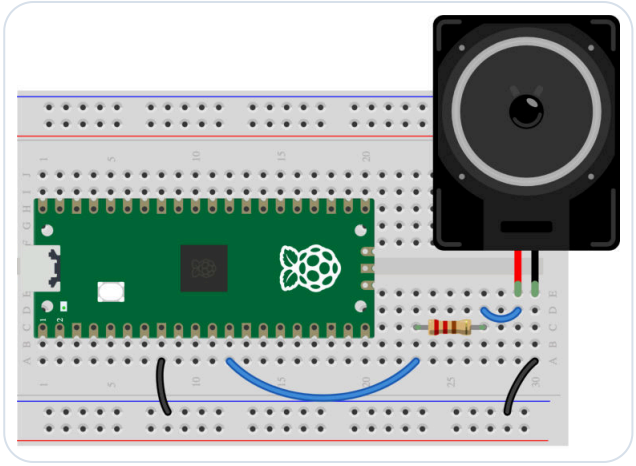


Pick another note from the menu, or change how long it plays.

Wiring

PICO



GP9 (pin 12) → 220 ohms → speaker red (+) lead; black lead → **GND** (pin 8) via the − rail

Motor driver

DO | PICO BLOCKS | OUTPUT | MOVEMENT

What does it do?

A motor driver lets a low-power GPIO control a motor that needs more current.

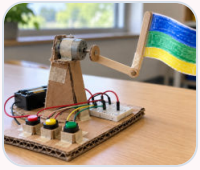
The L293D contains H-bridges that can send current through a motor in either direction.

Two input pins choose forwards, backwards or stop, while the chip handles the motor current.

Uses

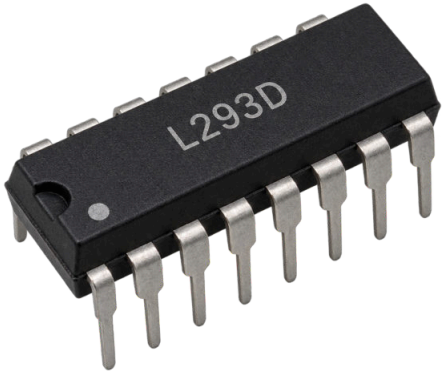
Electric car window

DETECT Read window switch → **DECIDE** Can it move safely? → **DO** Drive requested direction



Tiny flag waver

DETECT Count score-button presses → **DECIDE** Has the target been reached? → **DO** Wave the flag



Notch
Identifies pin 1

IN1 and IN2
Direction controls

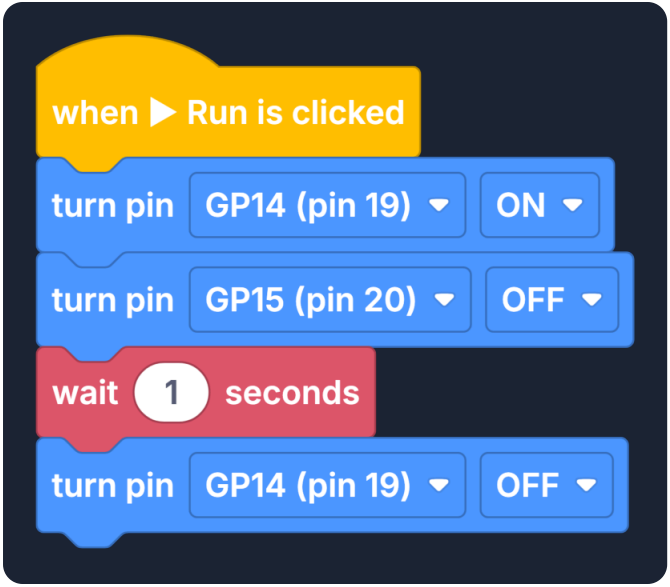
OUT1 and OUT2
Connect to the motor

Enable
Held high in this circuit

Blocks

MicroLab

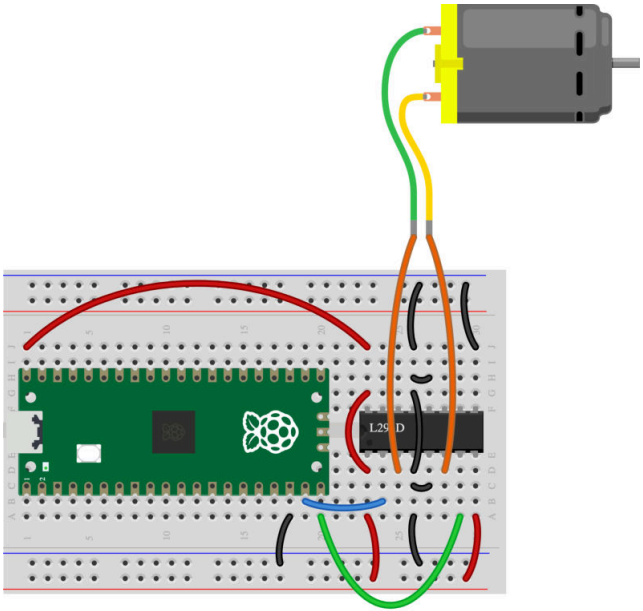
getmicrolab.com



GP14 ON and GP15 OFF drive forwards for one second. Swap them to go backwards.

Wiring

PICO



GP14 (pin 19) → IN1, **GP15** (pin 20) → IN2; OUT1 and OUT2 → motor; **VBUS** (pin 40), **GND** (pin 18)

Servo motor

DO | PICO BLOCKS | OUTPUT | MOVEMENT

What does it do?

A positional servo moves its arm, called a horn, to a chosen position and holds it there.

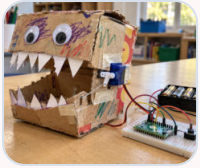
Short pulses on its signal wire set the position. The motor and position control are built into the case.

Use it for steering, pointing or opening a flap. A continuous-rotation servo controls speed instead; it cannot use this example to choose an angle.

Uses

Radio-controlled steering

DETECT: Read steering command → DECIDE: Which way should it turn? → DO: Set the steering position



Cardboard chomping monster

DETECT: Read feeding button → DECIDE: Is the monster hungry? → DO: Open its cardboard jaw

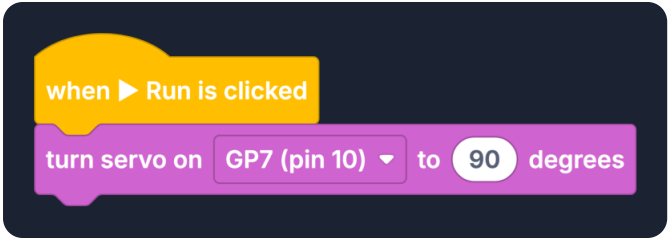
Horn	Signal	Power	Ground
Attach a light moving part	Usually orange, yellow or white	Usually red; about 5 V	Brown or black; join grounds

Raspberry Pi Foundation

Blocks

MicroLab

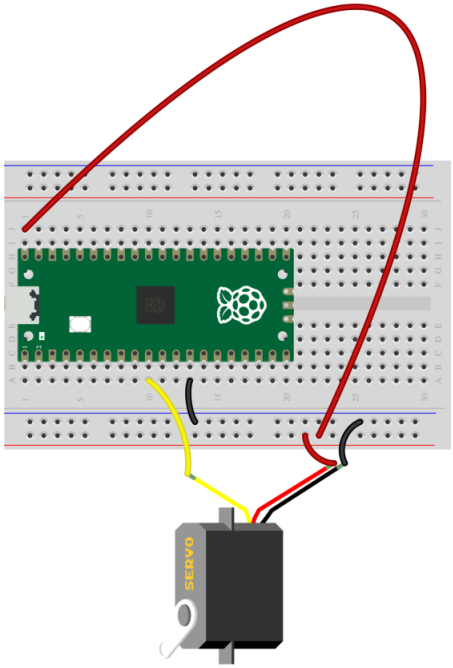
getmicrolab.com



Try 0 or 180 degrees to swing the arm to each end.

Wiring

PICO



GP7 (pin 10) → signal; **VBUS** (pin 40) → bottom + rail → servo red; **GND** (pin 13) → − rail → servo −. Small **SG90-size** servo only: USB powers it.

Button

DETECT | PICO | INPUT

What does it do?

A button lets someone tell your program to do something the moment they press.

Inside are two metal contacts. Pressing pushes them together so electricity can flow, and letting go springs them apart again.

Your projects can use it to start something, stop something, or count how many times it has been pressed.

Uses

Pedestrian crossing

DETECT: Button and traffic phase → DECIDE: Is it safe to cross? → DO: Change the lights

Do not press this

DETECT: Button is pressed → DECIDE: Is the trap armed? → DO: Play a silly sound

Cap
Press here

Legs
Four, joined in pairs

Gap
Straddle the channel

Raspberry Pi Foundation

Code

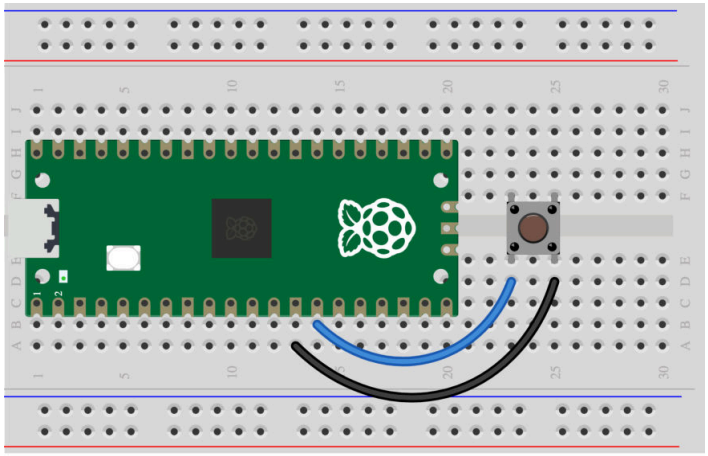
Raspberry Pi Pico

```
from picozero import Button

button = Button(10)

pressed = button.is_pressed
```

`pressed` is True or False. No resistor is needed — picozero switches on the pin's own pull-up. Also `button.when_pressed` and `button.when_released`.

Wiring

PICO



GP10 (pin 14) → button → **GND** (pin 13)

Capacitive touch

DETECT PICO INPUT

What does it do?

A capacitive touch sensor notices when a finger comes close to its metal pad.

It detects the tiny change in electric charge caused by your body, so no moving parts are needed.

Use it anywhere a project needs a hidden, wipe-clean or feather-light button.

Uses

Touch lamp

DETECT: Finger touches the pad → DECIDE: Is the lamp already on? → DO: Toggle the light

Banana piano

DETECT: A banana is touched → DECIDE: Which note should play? → DO: Play the note

Touch pad
The large metal spiral

SIG
Digital output

VCC and GND
3.3 V power

Raspberry Pi Foundation

Code

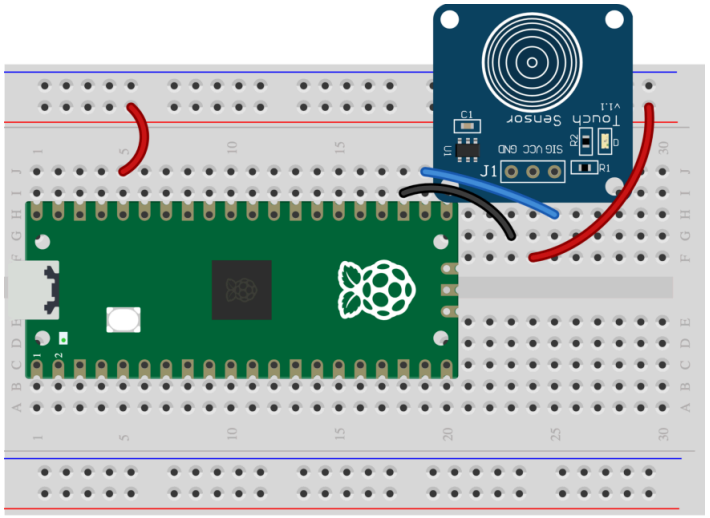
Raspberry Pi Pico

```
from picozero import TouchSensor

touch = TouchSensor(17)

touched = touch.is_touched
```

The TTP223 output goes high while touched. `touched` is True or False.

Wiring

PICO



GND (pin 23) → GND, **3V3 OUT** (pin 36) → VCC, **GP17** (pin 22) → SIG

LDR

DETECT PICO LIGHT

What does it do?

An LDR is a sensor that measures how bright or dark it is.

LDR stands for light-dependent resistor. Light shining on it lets electricity through more easily, so your code can tell light from dark.

Your projects can use it to react when a lamp comes on, when night falls, or when a shadow passes over.

Uses

Street light

DETECT: How dark it is → DECIDE: Is it darker than dusk? → DO: Switch the lamp on

Biscuit tin alarm

DETECT: Light inside the tin → DECIDE: Is the lid open? → DO: Sound the alarm

Window — Light lands here

Track — Zigzag sensor

Legs — Either way round

Code

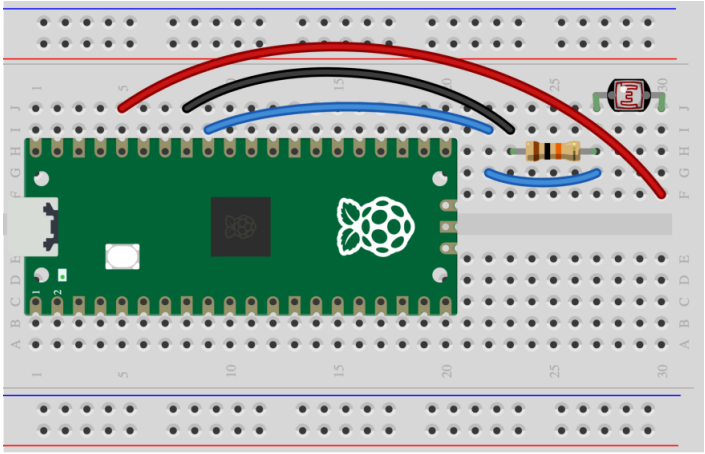
Raspberry Pi Pico

```
from picozero import AnalogInputDevice

ldr = AnalogInputDevice(27)

light = ldr.value
```

`light` runs from 0 in the dark to 1 in bright light.
Also `ldr.voltage`.

Wiring

PICO



3V3 (pin 36) → LDR → **GP27** (pin 32) → 10 kΩ → **GND** (pin 33)

Raspberry Pi Foundation

Line sensor

DETECT PICO INPUT

What does it do?

A line sensor tells light and dark surfaces apart underneath a moving project.

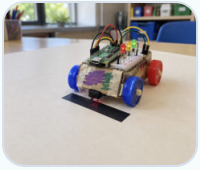
An infrared LED shines at the floor and a detector measures how much light bounces back.

Adjust the small potentiometer until its digital output changes reliably at the line edge.

Uses

Warehouse guide-line robot

DETECT Read guide-line position → **DECIDE** Is the vehicle drifting? → **DO** Correct the steering



Cardboard lap-counting car

DETECT Cross dark finish strip → **DECIDE** Is this a new lap? → **DO** Increase the lap count

Infrared pair
Faces the floor

Adjuster
Sets the threshold

S
Digital signal

Raspberry Pi Foundation

Code

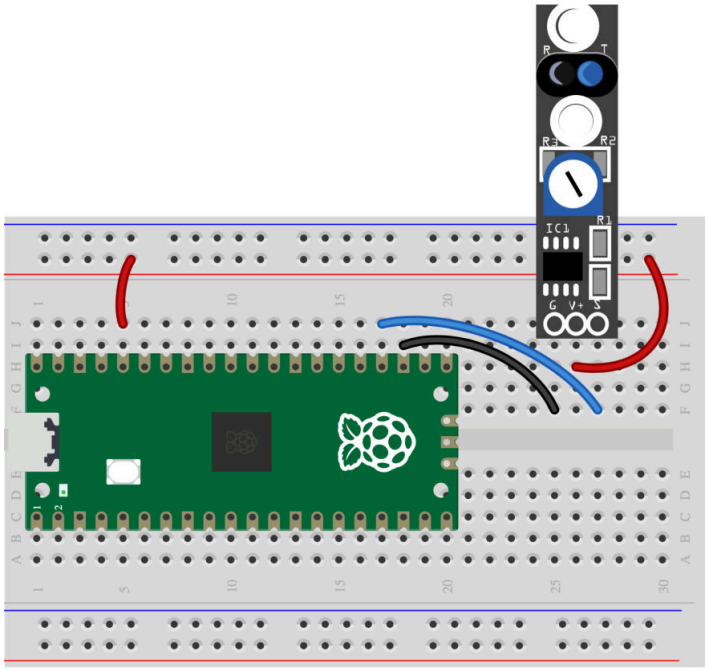
Raspberry Pi Pico

```
from picozero import DigitalInputDevice

line = DigitalInputDevice(18, pull_up=False)

on_line = line.is_active
```

Turn the blue adjuster until the module's output changes cleanly at the edge of the line.

Wiring

PICO



GND (pin 23) → G, **3V3 OUT** (pin 36) → V+, **GP18** (pin 24) → S

Distance sensor

DETECT PICO DISTANCE

What does it do?

An HC-SR04P distance sensor measures how far away something is without touching it.

It works like a bat. One side sends out a click too high for you to hear, the other side listens for the echo, and the time it takes gives the distance.

Your projects can use it to react as something comes closer, or to measure a gap that keeps changing.

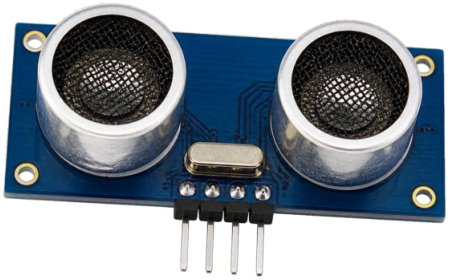
Uses

Parking sensor

DETECT: Gap to the wall → DECIDE: Is it closer than 30 cm? → DO: Beep faster

Air harp

DETECT: How far your hand is → DECIDE: Which note is that? → DO: Play the note



VCC	Trig	Echo	GND
Power, 3V3	Sends a click	Times the return	Ground

Raspberry Pi Foundation

Code

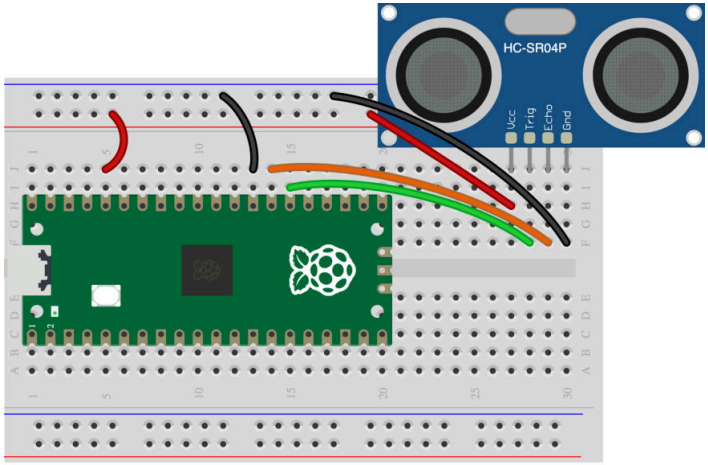
Raspberry Pi Pico

```
from picozero import DistanceSensor

sensor = DistanceSensor(
    echo=21, trigger=20
)

distance = sensor.distance
```

`distance` is in metres and stops at 1. Pass `max_distance=2` to see further, or read `sensor.value` for 0 to 1. Use an **HC-SR04P**, powered from **3V3**.

Wiring

PICO



HC-SR04P · Trig **GP20** (pin 26) · Echo **GP21** (pin 27) · **3V3 OUT** (pin 36) to top + rail · **GND** (pin 28) to top − rail

PIR

DETECT PICO MOTION

What does it do?

A PIR spots movement nearby, such as somebody walking past.

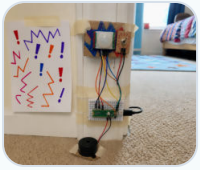
PIR stands for passive infrared. It watches the warmth that bodies give off and reports when that warmth moves, so it notices people and animals rather than furniture.

Your projects can use it to wake something up the moment somebody arrives, and go quiet again when they leave.

Uses

Burglar alarm

DETECT Movement in the room → **DECIDE** Should the room be empty? → **DO** Raise the alarm



Sibling alarm

DETECT Movement at the door → **DECIDE** Is privacy mode armed? → **DO** Sound the warning

Dome	VCC	OUT	GND
Wide-angle lens	Power, 5V	High on movement	Ground

Raspberry Pi Foundation

Code

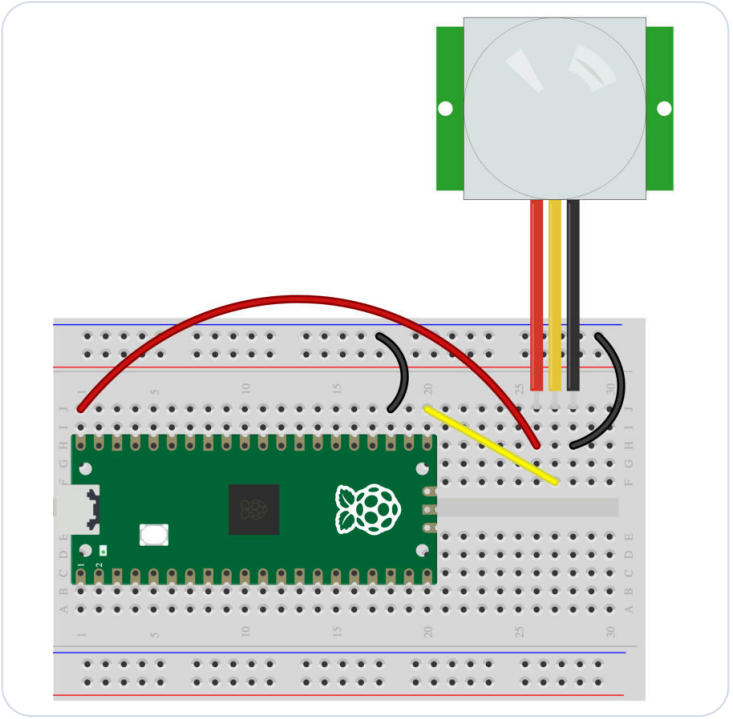
Raspberry Pi Pico

```
from picozero import MotionSensor

pir = MotionSensor(16)

moved = pir.motion_detected
```

`moved` is True or False. Also `pir.when_motion` and `pir.when_no_motion` to run a function the moment it changes. Most boards carry two dials that set how sensitive it is and how long OUT stays high.

Wiring

PICO



VBUS (pin 40) · OUT **GP16** (pin 21) · **GND** (pin 23)

AM312 PIR

DETECT | PICO | MOTION | MINI PIR

What does it do?

An **AM312** is a tiny PIR sensor that notices warm bodies moving nearby.

PIR means passive infrared. Its faceted lens divides the view into zones, so a person or animal moving between them makes the output switch on.

The complete AM312 module runs from 3V3 and uses very little power. Its sensitivity and roughly two-second output pulse are fixed.

Uses

Wardrobe light

DETECT Movement in the wardrobe → **DECIDE** Is someone reaching in? → **DO** Turn on the light

Dragon treasure alarm

DETECT Movement near the hoard → **DECIDE** Is someone sneaking closer? → **DO** Make the dragon roar

Lens	VCC	OUT	GND
Splits the view into zones	Power from 3V3	High on movement	Ground

Raspberry Pi Foundation

Code

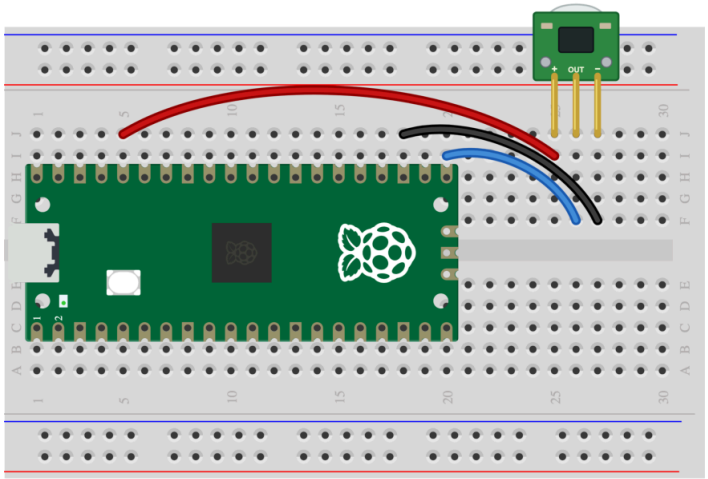
Raspberry Pi Pico

```
from picozero import MotionSensor

pir = MotionSensor(16)

moved = pir.motion_detected
```

`moved` is True or False. Use `pir.when_motion` and `pir.when_no_motion` to run functions when it changes. The AM312 has no sensitivity or delay dials; OUT normally stays high for about two seconds after movement.

Wiring

PICO



VCC → 3V3 OUT (pin 36); **OUT** → GP16 (pin 21); **GND** → GND (pin 23).

Potentiometer

DETECT PICO INPUT

What does it do?

A potentiometer lets your code read how far a knob has been turned.

This small version is called a trimpot. Turning the blue knob moves a contact called the wiper, changing the voltage your board reads.

Use it to choose a speed, a brightness or a sound. Turn gently and stop at the ends. The marking 103 means 10 kΩ.

Uses

Security-light adjustment

DETECT Read sensitivity setting → DECIDE Is movement strong enough? → DO Switch the light on

Alien voice tuner

DETECT Knob position → DECIDE Should it rumble or squeak? → DO Rumble or squeak

Knob	Outer pair	Wiper
Turn to adjust	Pins 1 and 3	Pin 2, offset

Raspberry Pi Foundation

Code

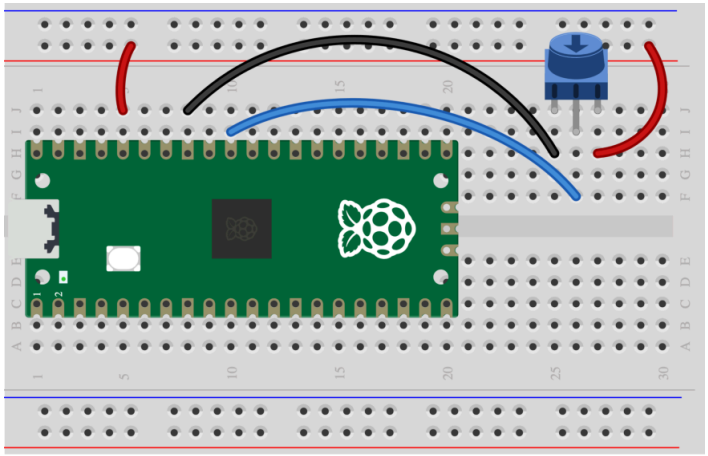
Raspberry Pi Pico

```
from picozero import Pot

pot = Pot(26)

setting = pot.value
```

`setting` runs from 0 to 1. Read `pot.value` again after turning; `pot.voltage` gives volts. Power the trimpot from **3V3**.

Wiring

PICO



Trimpot pin **1** → **GND** (pin 33) · **2** (wiper) → **GP26** (pin 31) · **3** → **3V3 OUT** (pin 36). Each pin has its own breadboard row.

Temperature sensor

DETECT | PICO | INPUT | ANALOGUE

What does it do?

A TMP36 turns temperature into a changing voltage that a program can measure.

Its output rises by 10 millivolts for every degree Celsius, with 0.5 V representing 0 °C.

The Pico reads that voltage directly; the Raspberry Pi uses an MCP3008 analogue-to-digital converter.

Uses

Refrigerator cooling

DETECT Measure fridge temperature → **DECIDE** Is it too warm? → **DO** Increase the cooling

Soup drama meter

DETECT Measure bowl surface → **DECIDE** Which drama level fits? → **DO** Move the drama pointer

Flat face	+Vs	Vout	GND
Use it to identify the pins	Left pin	Centre pin	Right pin

Raspberry Pi Foundation

Code

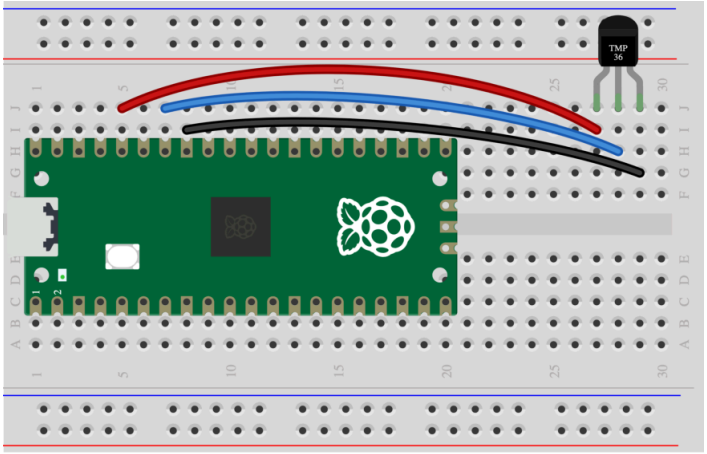
Raspberry Pi Pico

```
from picozero import TemperatureSensor

convert = lambda voltage: (voltage - 0.5) * 100
sensor = TemperatureSensor(28, conversion=convert)

temperature = sensor.temp
```

Read the TMP36 with its flat face towards you; the pins are +Vs, Vout, GND from left to right.

Wiring

PICO



TMP36 **+Vs** → 3V3 OUT (pin 36), **Vout** → GP28 (pin 34), **GND** → GND (pin 33)

Temperature probe

DETECT | PICO | DS18B20 | DIGITAL

What does it do?

A **DS18B20** measures temperature and sends a digital reading in degrees Celsius.

This probe has a sealed metal tip on a cable with three wires. Check the supplier's pinout; wire colours can vary.

Only the sealed probe goes in water. Keep the board and all wire connections dry.

Uses

Aquarium temperature alarm

DETECT Measure tank temperature → **DECIDE** Is it too cold? → **DO** Show a warning

Penguin bath alarm

DETECT Measure bath temperature → **DECIDE** Is it too warm? → **DO** Light a warning LED

VDD	DQ	GND	4.7 kΩ
Power from 3V3	Digital data	Ground	Resistor from DQ to 3V3

Raspberry Pi Foundation

Code

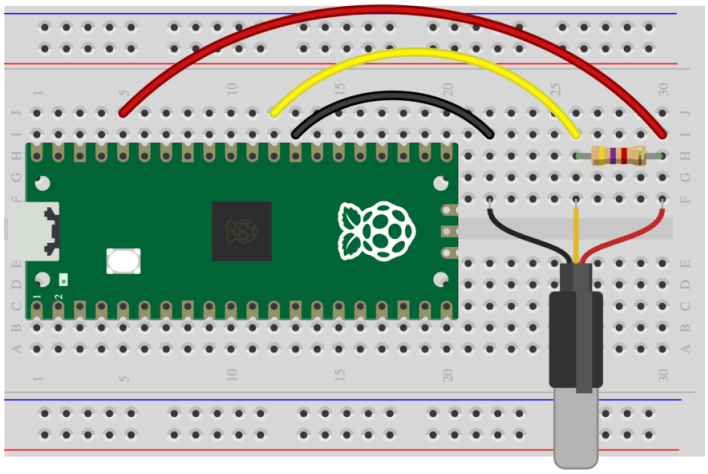
Raspberry Pi Pico

```
from machine import Pin
import onewire, ds18x20
from time import sleep_ms

bus = onewire.OneWire(Pin(22))
sensor = ds18x20.DS18X20(bus)
probes = sensor.scan()
assert probes, "Check probe wiring"
sensor.convert_temp()
sleep_ms(750)
temperature = sensor.read_temp(probes[0])
print(temperature)
```

Repeat the conversion and 750 ms wait for each new reading.

Wiring

PICO



VDD → 3V3 OUT (pin 36); **DQ** → GP22 (pin 29); **GND** → GND (pin 28). **4.7 kΩ** between DQ and 3V3.

Tilt switch

DETECT PICO INPUT

What does it do?

A tilt switch changes state when it tips past a particular angle.

While it stands upright, a tiny metal ball rests on two contacts and completes the circuit. Tip it over and the ball rolls away, breaking the circuit.

It can detect that an object has been picked up, tipped over or opened.

Uses

Iron safety shut-off

DETECT: Iron tips sideways → DECIDE: Has the delay elapsed? → DO: Switch the heater off

Grumpy pencil case

DETECT: Case is tipped → DECIDE: Has it stayed tipped? → DO: Make a grumpy noise

Metal can
Contains the rolling ball

Legs
No polarity

Angle
Changes the contact

Raspberry Pi Foundation

Code

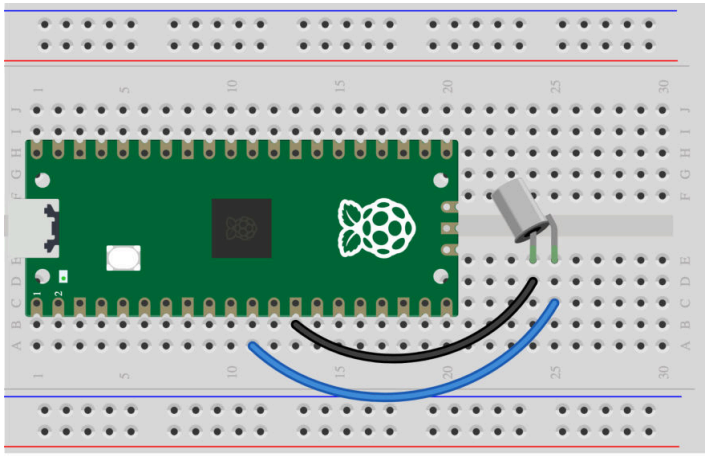
Raspberry Pi Pico

```
from picozero import Switch

tilt = Switch(8)

tilted = tilt.is_open
```

> The reading may flicker while the ball rolls. Check it more than once before treating it as a definite tilt.

Wiring

PICO



GP8 (pin 11) → tilt switch → **GND** (pin 13)

Raspberry Pi Foundation

Toggle switch

DETECT | PICO | INPUT

What does it do?

A toggle switch stays in the position you move it to, unlike a push button.

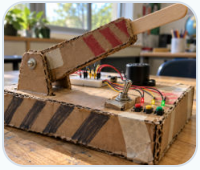
Its centre COM leg connects to one outside leg in each position.

It is useful when a project needs a lasting on/off choice or two modes.

Uses

Auto/manual selector

DETECT Read selector position → **DECIDE** Is automatic mode selected? → **DO** Run the selected mode



Serious business lever

DETECT Lever and check buttons → **DECIDE** Are all checks complete? → **DO** Start unnecessary sirens

Lever Stays where it is moved

COM Centre pin

L1 and L2 The two choices

Raspberry Pi Foundation

Code

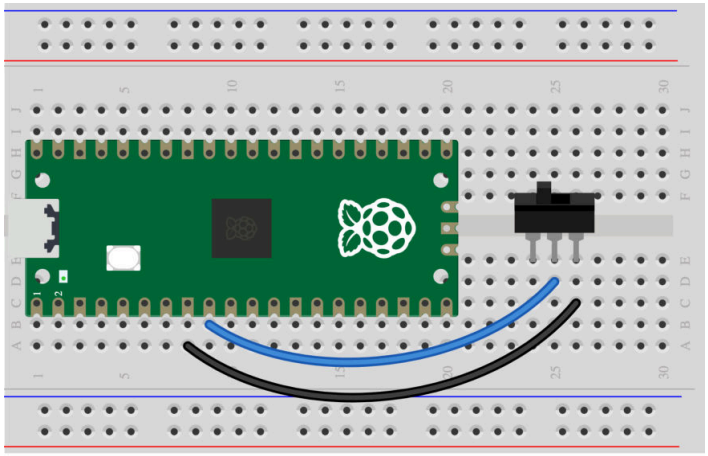
Raspberry Pi Pico

```
from picozero import Switch

mode = Switch(6)

mode_two = mode.is_closed
```

Connect COM and one outside pin. The unused outside pin can be left unconnected.

Wiring

PICO



GP6 (pin 9) → COM; right-hand outside pin → **GND** (pin 8)

LED

DO PICO LIGHT

What does it do?

An LED is a tiny light that your code can switch on, switch off, or blink.

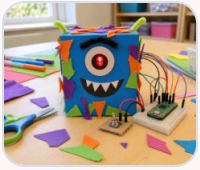
LED stands for light-emitting diode. It glows when electricity flows the right way through it, so the long leg and the short leg matter.

Your projects can use LEDs to show a warning, celebrate something, or tell people what is happening.

Uses

Car brake light

DETECT Brake control signal → **DECIDE** Is braking requested? → **DO** Light the red LEDs



One-eyed monster box

DETECT Box lid is opened → **DECIDE** Is guard mode armed? → **DO** Glow the monster eye

Anode
Long leg, to GPIO

Cathode
Short leg, to GND

Lens
Glows here

Raspberry Pi Foundation

Code

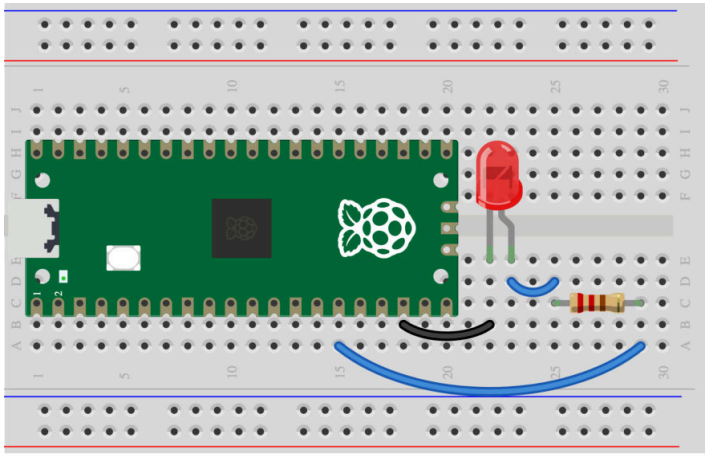
Raspberry Pi Pico

```
from picozero import LED

led = LED(11)

led.on()
```

Also `led.off()`, `led.toggle()` and `led.blink()`. Give `blink()` an `on_time` and an `off_time` to set the flash speed.

Wiring

PICO



GP11 (pin 15) → 220 Ω → LED → **GND** (pin 18)

Raspberry Pi Foundation

RGB LED

DO PICO OUTPUT LIGHT

What does it do?

An RGB LED puts red, green and blue lights inside one clear package.

Varying the brightness of the three colours mixes millions of possible colours.

A common-cathode LED has one shared negative leg and needs one resistor for each colour.

Uses

Wi-Fi router status

DETECT Read connection status → **DECIDE** Which status colour is needed? → **DO** Mix the LED colour

Mood crystal

DETECT Read the mood dial → **DECIDE** What is the crystal's mood? → **DO** Glow that colour

Clear lens
Contains three LEDs

Long leg
Common cathode

R, G and B legs
One 220 ohm resistor each

Raspberry Pi Foundation

Code

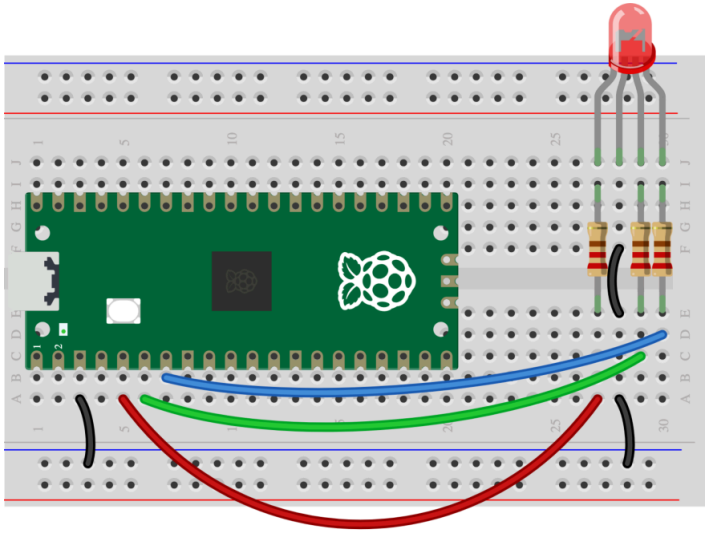
Raspberry Pi Pico

```
from picozero import RGBLED

light = RGBLED(red=3, green=4, blue=5)

light.color = (255, 0, 255)
```

picozero colour values run from 0 to 255. This example makes magenta.

Wiring

PICO



R → **GP3** (pin 5), G → **GP4** (pin 6), B → **GP5** (pin 7); common → **GND** (pin 3)

NeoPixels

DO | PICO | OUTPUT | LIGHT

What does it do?

A NeoPixel is a full-colour LED with a tiny controller built into it.

One data wire carries colour instructions, and each pixel passes the remaining instructions onwards.

The 3.3 V-compatible JST breakout shown here can connect directly to a Pi or Pico.

Uses

Gaming PC lighting

DETECT Read lighting selection → **DECIDE** Which colour should each pixel be? → **DO** Light individual pixels

Disco shoelace meter

DETECT Read shoe movement → **DECIDE** Which light pattern should run? → **DO** Animate the pixels

Pixel RGB LED and controller

IN Data from the board

OUT Data to another pixel

VDD and GND 3.3 V power

Raspberry Pi Foundation

Code

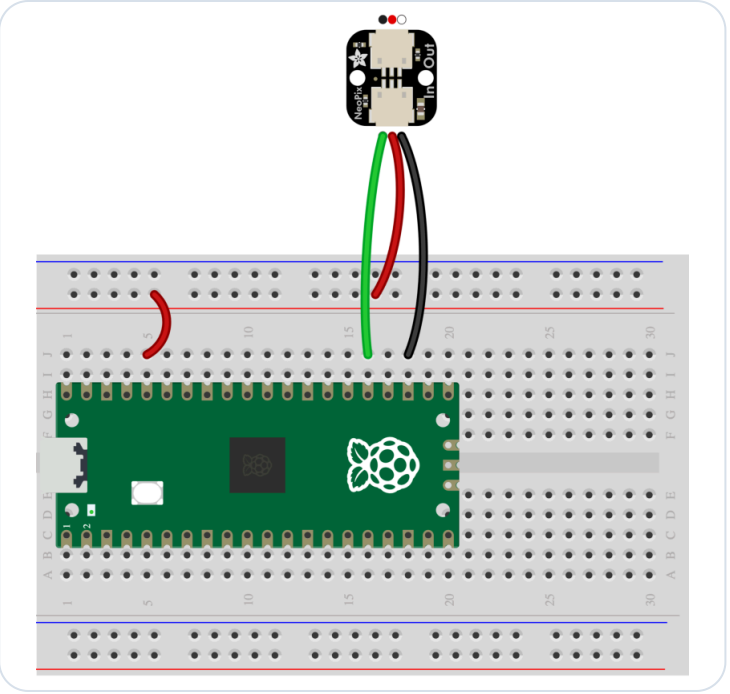
Raspberry Pi Pico

```
from machine import Pin
from neopixel import NeoPixel

pixels = NeoPixel(Pin(19), 1)
pixels[0] = (255, 0, 80)
pixels.write()
```

This wiring is for the pictured 3.3 V-compatible breakout. Generic 5 V strips usually need a separate supply and a logic-level shifter.

Wiring

PICO



GP19 (pin 25) → IN, **3V3 OUT** (pin 36) → top + rail → VDD, **GND** (pin 23) → GND

Active buzzer

DO PICO SOUND

What does it do?

An active buzzer makes one built-in tone. Switch it on and off for warning beeps or rhythms; its pitch stays the same.

Recognise it: check the packet for **active** or **built-in oscillator**. Shape, stickers and + marks do not prove the type.

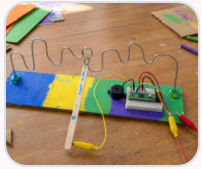
Use a labelled **3.3 V GPIO-compatible** buzzer.

Uses

Smoke alarm

One warning tone is enough; no changing notes needed.

DETECT Measure smoke level → **DECIDE** Is smoke above the limit? → **DO** Repeat one warning tone



Buzz wire game

Choose active: one fixed buzz clearly signals failure.

DETECT Loop touches wire → **DECIDE** Is this the third strike? → **DO** Buzz at one pitch

+ leg Marked +, to GPIO

– leg Other leg, to GND

Vent Sound comes out

Raspberry Pi Foundation

Code

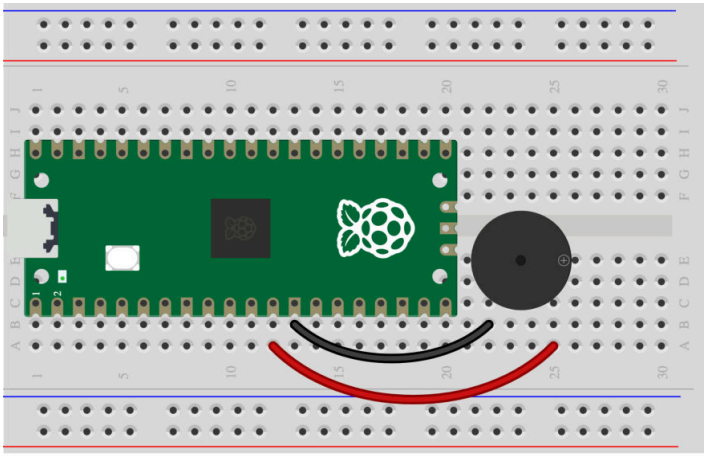
Raspberry Pi Pico

```
from picozero import Buzzer

buzzer = Buzzer(9)

buzzer.on()
```

Also `buzzer.off()`, `buzzer.toggle()` and `buzzer.beep()`. Give `beep()` an `on_time` and an `off_time` to set the rhythm. Faster beeps do not mean a higher pitch.

Wiring

PICO



GP9 (pin 12) → **+** buzzer **–** → **GND** (pin 13)

Passive buzzer

DO PICO SOUND PWM

What does it do?

A passive buzzer needs fast pulses (PWM) to make sound. Changing their frequency changes its pitch: low or high notes.

Recognise it: check for **passive**, **external drive** or **no oscillator** on the packet. Both types can look identical; + only marks polarity.

Use a small **3.3 V GPIO-compatible piezo** sounder.

Uses

Washing-machine chime

PWM plays tunes; active buzzers have only one pitch.

DETECT Read cycle status → **DECIDE** Has the wash finished? → **DO** Play a short tune

Robot voice box

Change pitch for happy chirps or grumpy growls.

DETECT Read mood button → **DECIDE** Is happy pressed? → **DO** Chirp high or low

+ leg Marked +, to GPIO

− leg Other leg, to GND

Vent Sound comes out

Raspberry Pi Foundation

Code

rpf.io/soundfx

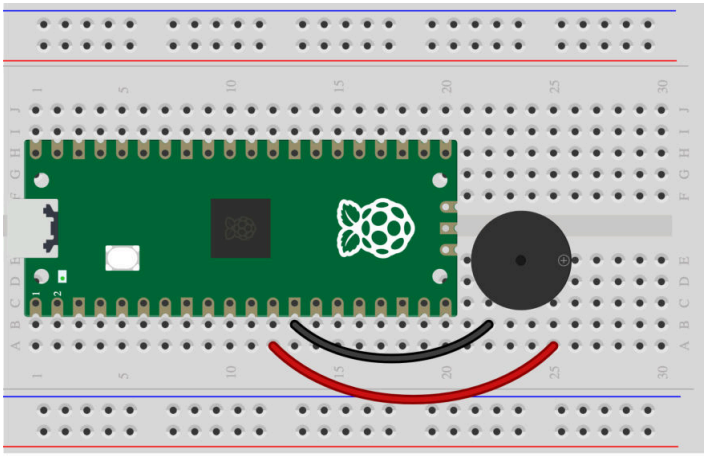
Raspberry Pi Pico

```
from picozero import PWMBuzzer

# GP9, 2000 Hz, 50% duty
buzzer = PWMBuzzer(9, 2000, 32768)

buzzer.on()
```

Try `buzzer.freq = 1000`; silence with `buzzer.off()`. Higher frequency means higher pitch. Beep speed is separate; both types can make rhythms.

Wiring

PICO



GP9 (pin 12) → **+** buzzer **–** → **GND** (pin 13)

Speaker

DO PICO OUTPUT SOUND

What does it do?

A small passive speaker turns a changing electrical signal into sound.

Current through its coil moves a cone backwards and forwards, pushing the air.

The 220 ohm resistor shown here protects the GPIO; it also keeps this simple circuit fairly quiet.

Uses

Door chime

DETECT Button and chime setting → **DECIDE** Which chime should play? → **DO** Play the notes

Tiny fanfare machine

DETECT Count achievement-button presses → **DECIDE** Has the target been reached? → **DO** Play triumphant notes

Cone
Pushes the air

Coil and magnet
Create movement

Terminals
Connect through 220 ohms

Raspberry Pi Foundation

Code

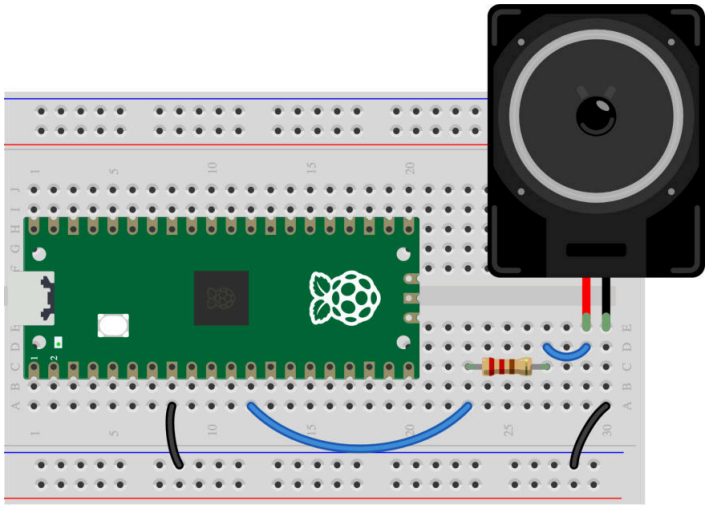
Raspberry Pi Pico

```
from picozero import Speaker

speaker = Speaker(9)

speaker.play("E4", duration=1)
```

For a louder speaker, use a transistor or amplifier rather than removing the protective resistor.

Wiring

PICO



GP9 (pin 12) → 220 ohms → speaker red (+) lead; black lead → **GND** (pin 8) via the − rail

Motor driver

DO | PICO | OUTPUT | MOVEMENT

What does it do?

A motor driver lets a low-power GPIO control a motor that needs more current.

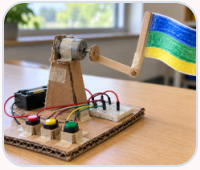
The L293D contains H-bridges that can send current through a motor in either direction.

Two input pins choose forwards, backwards or stop, while the chip handles the motor current.

Uses

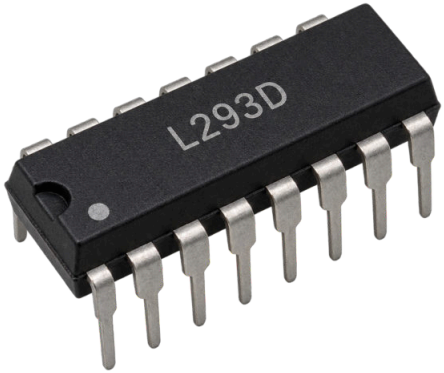
Electric car window

DETECT Read window switch → **DECIDE** Can it move safely? → **DO** Drive requested direction



Tiny flag waver

DETECT Count score-button presses → **DECIDE** Has the target been reached? → **DO** Wave the flag



Notch
Identifies pin 1

IN1 and IN2
Direction controls

OUT1 and OUT2
Connect to the motor

Enable
Held high in this circuit

Raspberry Pi Foundation

Code

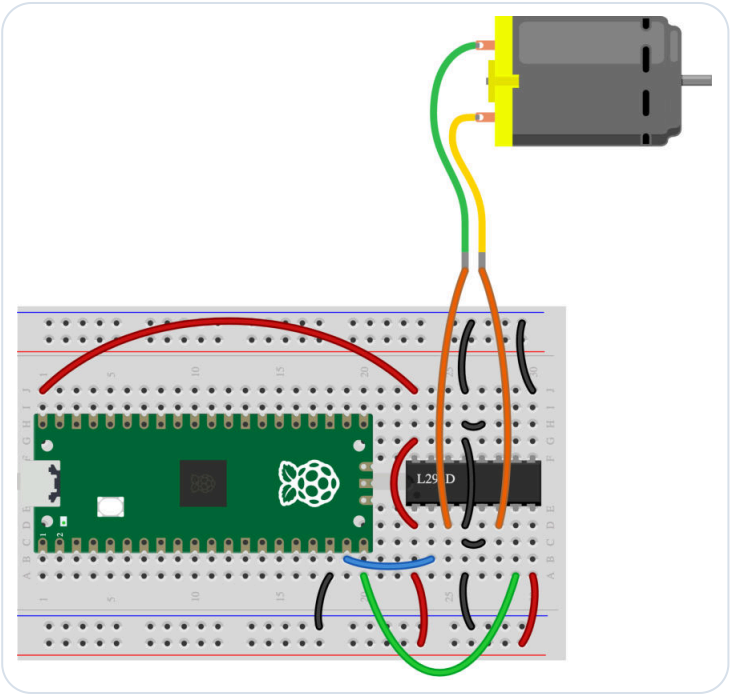
Raspberry Pi Pico

```
from picozero import Motor

motor = Motor(14, 15)

motor.forward()
```

This card powers one small 5 V motor from the board's 5 V rail. Use a separate motor supply for larger motors, with the grounds joined.

Wiring

PICO



GP14 (pin 19) → IN1, **GP15** (pin 20) → IN2; OUT1 and OUT2 → motor; **VBUS** (pin 40), **GND** (pin 18)

Servo motor

DO | PICO | OUTPUT | MOVEMENT

What does it do?

A positional servo moves its arm, called a horn, to a chosen position and holds it there.

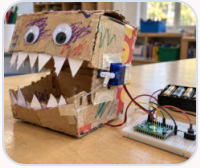
Short pulses on its signal wire set the position. The motor and position control are built into the case.

Use it for steering, pointing or opening a flap. A continuous-rotation servo controls speed instead; it cannot use this example to choose an angle.

Uses

Radio-controlled steering

DETECT Read steering command → **DECIDE** Which way should it turn? → **DO** Set the steering position



Cardboard chomping monster

DETECT Read feeding button → **DECIDE** Is the monster hungry? → **DO** Open its cardboard jaw

Horn	Signal	Power	Ground
Attach a light moving part	Usually orange, yellow or white	Usually red; about 5 V	Brown or black; join grounds

Raspberry Pi Foundation

Code

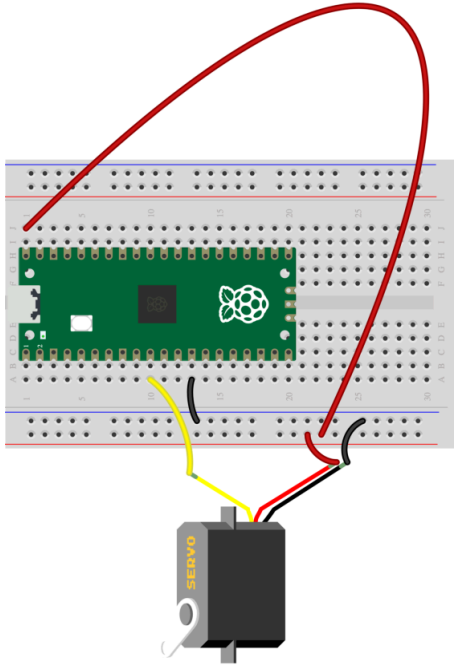
Raspberry Pi Pico

```
from picozero import Servo
from time import sleep

servo = Servo(7)
servo.mid()
sleep(2)
```

Test unloaded: the horn centres. Try min() or max() instead of mid(). Travel varies; stop if it binds or buzzes. Check lead order.

Wiring

PICO



GP7 (pin 10) → signal; **VBUS** (pin 40) → bottom + rail → servo red; **GND** (pin 13) → − rail → servo −. Small **SG90-size** servo only: USB powers it.

Stepper motor

DO | PICO | OUTPUT | MOVEMENT

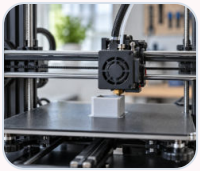
What does it do?

A stepper motor turns in small, repeatable steps instead of spinning freely.

Energising its coils in sequence moves the shaft one step at a time and holds its position.

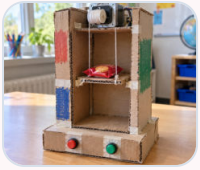
The ULN2003 board supplies the coil current for the common 28BYJ-48 motor shown here.

Uses



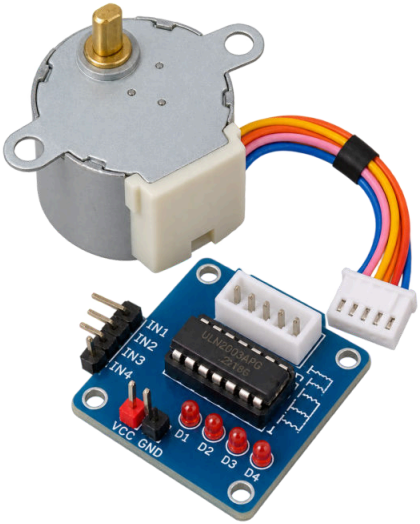
3D printer positioning

DETECT Read next print position → **DECIDE** How many steps are needed? → **DO** Move the print head



Snack lift

DETECT Snack request arrives → **DECIDE** Which floor needs it? → **DO** Step lift into place



Output shaft Moves in measured steps

Five-wire plug Connects to the driver

IN1 to IN4 Coil sequence

5V and GND Driver power

Raspberry Pi Foundation

Code

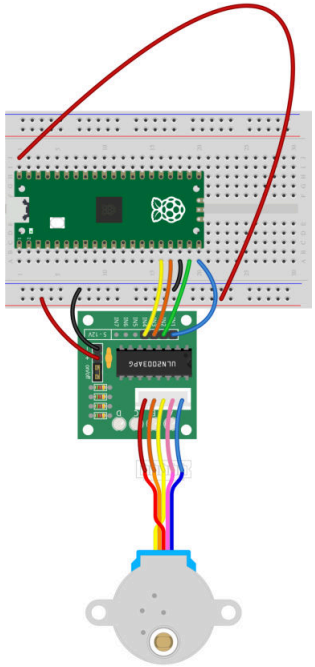
Raspberry Pi Pico

```
from picozero import Stepper

motor = Stepper((15, 14, 13, 12))

motor.step(100)
```

Call `motor.step(100)` again to keep moving. Add `direction="ccw"` to turn the other way.

Wiring

PICO



IN1-IN4 → GP **15, 14, 13, 12** (pins 20, 19, 17, 16); driver → **VBUS** (pin 40, bottom + rail) and **GND** (pin 18)

Button

DETECT | RASPBERRY PI | INPUT

What does it do?

A button lets someone tell your program to do something the moment they press.

Inside are two metal contacts. Pressing pushes them together so electricity can flow, and letting go springs them apart again.

Your projects can use it to start something, stop something, or count how many times it has been pressed.

Uses

Pedestrian crossing

DETECT Button and traffic phase → **DECIDE** Is it safe to cross? → **DO** Change the lights

Do not press this

DETECT Button is pressed → **DECIDE** Is the trap armed? → **DO** Play a silly sound

Cap
Press here

Legs
Four, joined in pairs

Gap
Straddle the channel

Raspberry Pi Foundation

Code

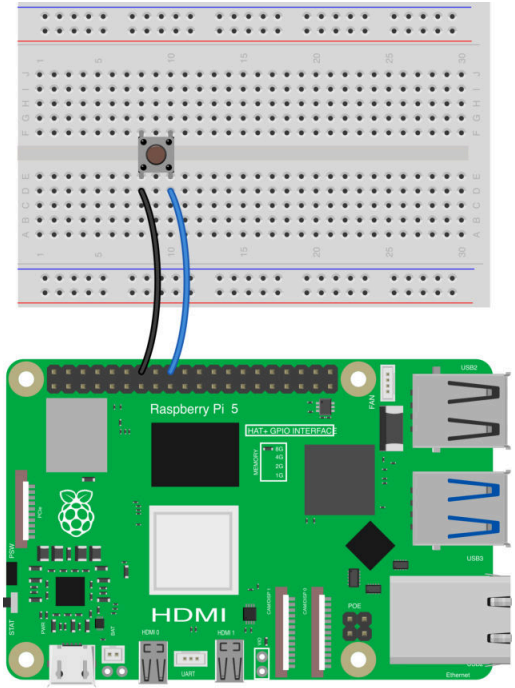
Raspberry Pi

```
from gpiozero import Button

button = Button(24)

pressed = button.is_pressed
```

`pressed` is True or False. No resistor is needed — gpiozero switches on the pin's own pull-up. Also `button.when_pressed` and `button.when_released`.

Wiring

RASPBERRY PI 5



GPIO 24 (pin 18) → button → **GND** (pin 14)

Capacitive touch

DETECT | RASPBERRY PI | INPUT

What does it do?

A capacitive touch sensor notices when a finger comes close to its metal pad.

It detects the tiny change in electric charge caused by your body, so no moving parts are needed.

Use it anywhere a project needs a hidden, wipe-clean or feather-light button.

Uses

Touch lamp

DETECT: Finger touches the pad → DECIDE: Is the lamp already on? → DO: Toggle the light

Banana piano

DETECT: A banana is touched → DECIDE: Which note should play? → DO: Play the note

Touch pad
The large metal spiral

SIG
Digital output

VCC and GND
3.3 V power

Raspberry Pi Foundation

Code

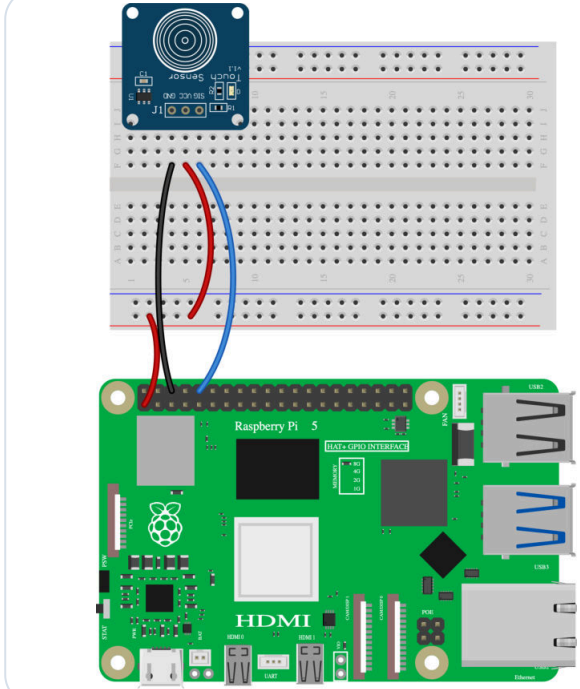
Raspberry Pi

```
from gpiozero import DigitalInputDevice

touch = DigitalInputDevice(15, pull_up=False)

touched = touch.is_active
```

The TTP223 output goes high while touched. `touched` is True or False.

Wiring

RASPBERRY PI 5



GND (pin 6) → GND, **3V3** (pin 1) → VCC, **GPIO 15** (pin 10) → SIG

LDR

DETECT | RASPBERRY PI | LIGHT

What does it do?

An LDR is a sensor that measures how bright or dark it is.

LDR stands for light-dependent resistor. Light shining on it lets electricity through more easily, so your code can tell light from dark.

Your projects can use it to react when a lamp comes on, when night falls, or when a shadow passes over.

Uses

Street light

DETECT How dark it is → **DECIDE** Is it darker than dusk? → **DO** Switch the lamp on

Biscuit tin alarm

DETECT Light inside the tin → **DECIDE** Is the lid open? → **DO** Sound the alarm

Window Light lands here

Track Zigzag sensor

Legs Either way round

Raspberry Pi Foundation

Code

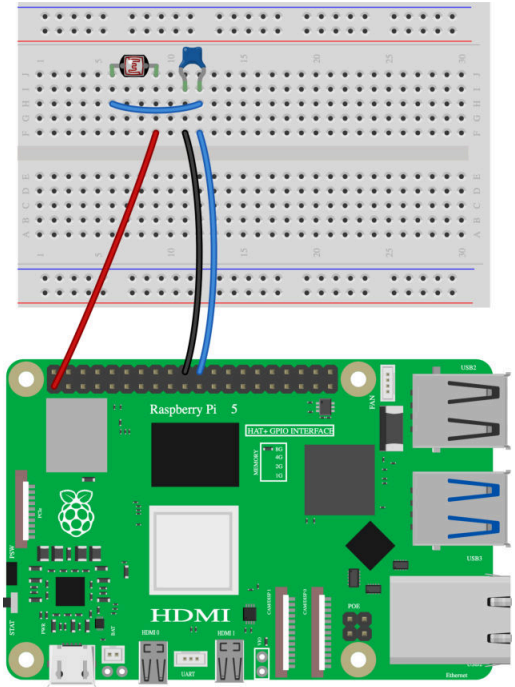
Raspberry Pi

```
from gpiozero import LightSensor

ldr = LightSensor(25)

light = ldr.value
```

`light` runs from 0 in the dark to 1 in bright light. Also `ldr.light_detected` and `ldr.when_light`.

Wiring

RASPBERRY PI 5



3V3 (pin 1) → LDR → **GPIO 25** (pin 22) → 1 µF → **GND** (pin 20)

Line sensor

DETECT | RASPBERRY PI | INPUT

What does it do?

A line sensor tells light and dark surfaces apart underneath a moving project.

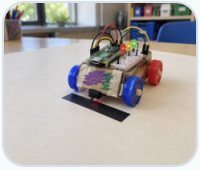
An infrared LED shines at the floor and a detector measures how much light bounces back.

Adjust the small potentiometer until its digital output changes reliably at the line edge.

Uses

Warehouse guide-line robot

DETECT Read guide-line position → **DECIDE** Is the vehicle drifting? → **DO** Correct the steering



Cardboard lap-counting car

DETECT Cross dark finish strip → **DECIDE** Is this a new lap? → **DO** Increase the lap count

Infrared pair
Faces the floor

Adjuster
Sets the threshold

S
Digital signal

Raspberry Pi Foundation

Code

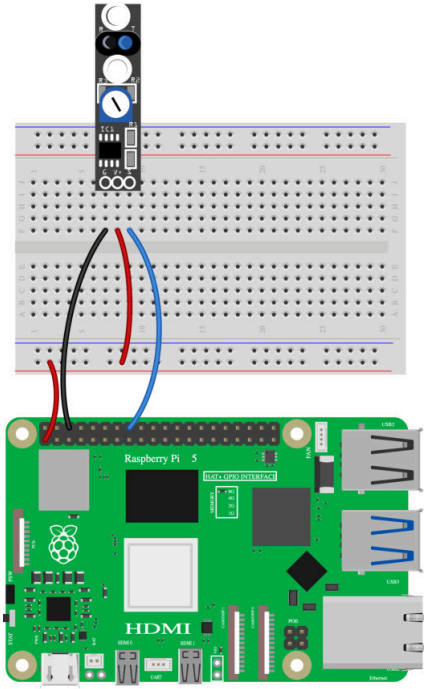
Raspberry Pi

```
from gpiozero import LineSensor

line = LineSensor(23)

on_line = line.is_active
```

> Turn the blue adjuster until the module's output changes cleanly at the edge of the line.

Wiring

RASPBERRY PI 5



GND (pin 6) → G, **3V3** (pin 1) → V+, **GPIO 23** (pin 16) → S

Distance sensor

DETECT | RASPBERRY PI | DISTANCE

What does it do?

An HC-SR04P distance sensor measures how far away something is without touching it.

It works like a bat. One side sends out a click too high for you to hear, the other side listens for the echo, and the time it takes gives the distance.

Your projects can use it to react as something comes closer, or to measure a gap that keeps changing.

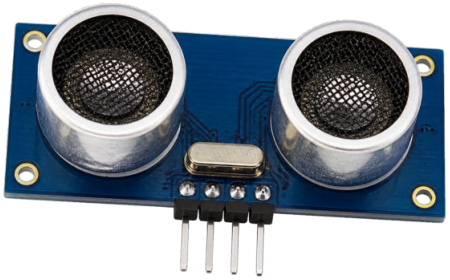
Uses

Parking sensor

DETECT: Gap to the wall → DECIDE: Is it closer than 30 cm? → DO: Beep faster

Air harp

DETECT: How far your hand is → DECIDE: Which note is that? → DO: Play the note



VCC	Trig	Echo	GND
Power, 3V3	Sends a click	Times the return	Ground

Raspberry Pi Foundation

Code

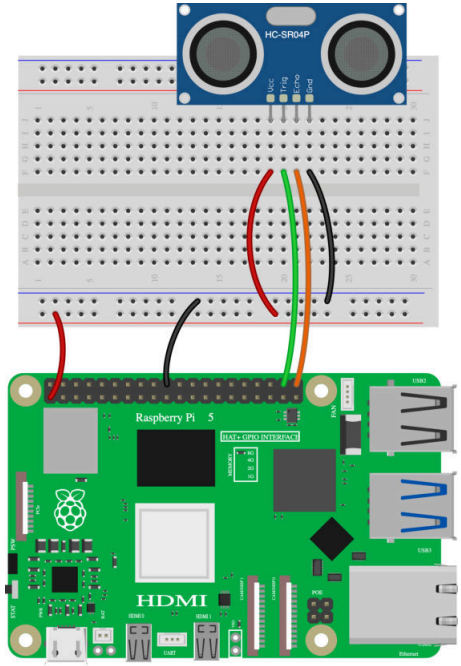
Raspberry Pi

```
from gpiozero import DistanceSensor

sensor = DistanceSensor(
    echo=21, trigger=20
)

distance = sensor.distance
```

`distance` is in metres and stops at 1. Pass `max_distance=2` to see further, or read `sensor.value` for 0 to 1. Use an **HC-SR04P**, powered from **3V3**.

Wiring

RASPBERRY PI 5



HC-SR04P · **3V3** (pin 1) to + rail · Trig **GPIO 20** (pin 38) · Echo **GPIO 21** (pin 40) · **GND** (pin 20) to − rail

PIR

DETECT | RASPBERRY PI | MOTION

What does it do?

A PIR spots movement nearby, such as somebody walking past.

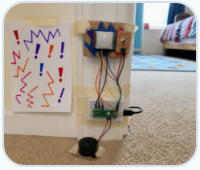
PIR stands for passive infrared. It watches the warmth that bodies give off and reports when that warmth moves, so it notices people and animals rather than furniture.

Your projects can use it to wake something up the moment somebody arrives, and go quiet again when they leave.

Uses

Burglar alarm

DETECT Movement in the room → **DECIDE** Should the room be empty? → **DO** Raise the alarm



Sibling alarm

DETECT Movement at the door → **DECIDE** Is privacy mode armed? → **DO** Sound the warning

Dome	VCC	OUT	GND
Wide-angle lens	Power, 5V	High on movement	Ground

Raspberry Pi Foundation

Code

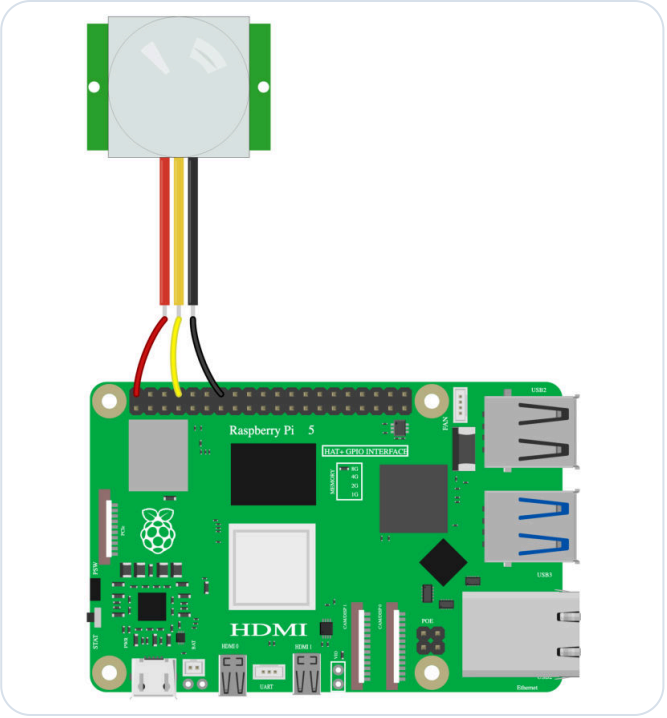
Raspberry Pi

```
from gpiozero import MotionSensor

pir = MotionSensor(14)

moved = pir.motion_detected
```

`moved` is True or False. Also `pir.when_motion` and `pir.when_no_motion` to run a function the moment it changes. Most boards carry two dials that set how sensitive it is and how long OUT stays high.

Wiring

RASPBERRY PI 5



5V (pin 2) · OUT **GPIO 14** (pin 8) · **GND** (pin 14)

AM312 PIR

DETECT | RASPBERRY PI | MOTION | MINI PIR

What does it do?

An **AM312** is a tiny PIR sensor that notices warm bodies moving nearby.

PIR means passive infrared. Its faceted lens divides the view into zones, so a person or animal moving between them makes the output switch on.

The complete AM312 module runs from 3V3 and uses very little power. Its sensitivity and roughly two-second output pulse are fixed.

Uses

Wardrobe light

DETECT Movement in the wardrobe → **DECIDE** Is someone reaching in? → **DO** Turn on the light

Dragon treasure alarm

DETECT Movement near the hoard → **DECIDE** Is someone sneaking closer? → **DO** Make the dragon roar

Lens	VCC	OUT	GND
Splits the view into zones	Power from 3V3	High on movement	Ground

Raspberry Pi Foundation

Code

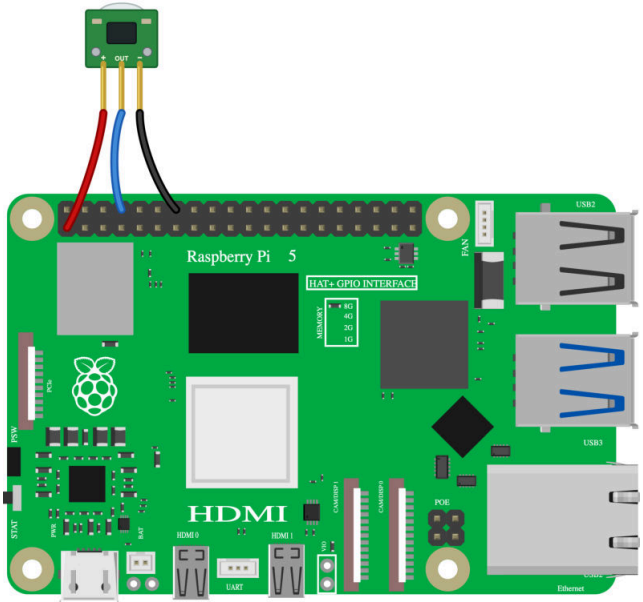
Raspberry Pi

```
from gpiozero import MotionSensor

pir = MotionSensor(14)

moved = pir.motion_detected
```

`moved` is True or False. Use `pir.when_motion` and `pir.when_no_motion` to run functions when it changes. The AM312 has no sensitivity or delay dials; OUT normally stays high for about two seconds after movement.

Wiring

RASPBERRY PI 5



VCC → 3V3 (pin 1); **OUT** → GPIO14 (pin 8); **GND** → GND (pin 14).

Potentiometer

DETECT | RASPBERRY PI | INPUT

What does it do?

A potentiometer lets your code read how far a knob has been turned.

This small version is called a trimpot. Turning the blue knob moves a contact called the wiper, changing the voltage your board reads.

Use it to choose a speed, a brightness or a sound. Turn gently and stop at the ends. The marking 103 means 10 kΩ.

Uses

Security-light adjustment

DETECT: Read sensitivity setting → DECIDE: Is movement strong enough? → DO: Switch the light on

Alien voice tuner

DETECT: Knob position → DECIDE: Should it rumble or squeak? → DO: Rumble or squeak

Knob	Outer pair	Wiper
Turn to adjust	Pins 1 and 3	Pin 2, offset

Code

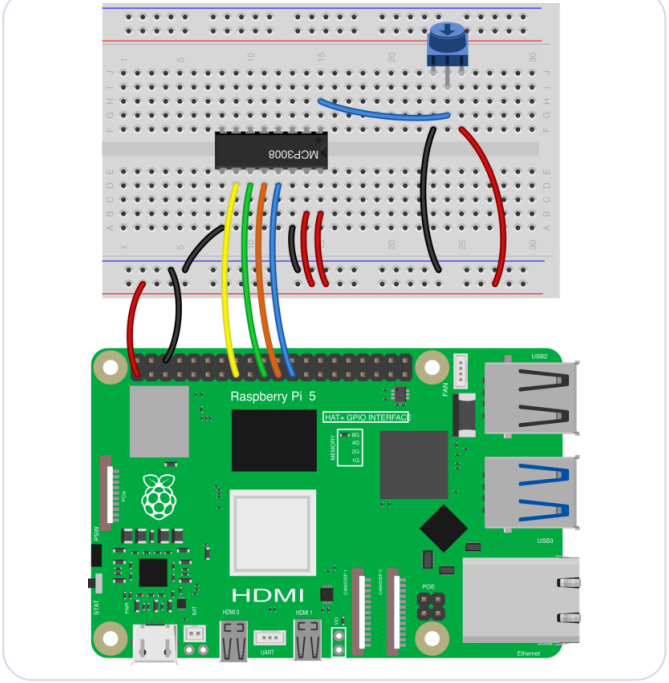
Raspberry Pi

```
from gpiozero import MCP3008

pot = MCP3008(0, select_pin=22)

setting = pot.value
```

`setting` runs from 0 to 1. Read `pot.value` again after turning; `pot.voltage` gives volts. The Pi needs the MCP3008 converter shown. Power the trimpot and converter from **3V3**.

Wiring

PI 5 + MCP3008



Wiper → **CH0**. CLK / DIN / DOUT / CS → **GPIO 11 / 10 / 9 / 22** (pins 23 / 19 / 21 / 15). **3V3** (pin 1), **GND** (pin 6). Chip notch faces right.

Temperature sensor

DETECT | RASPBERRY PI | INPUT | ANALOGUE

What does it do?

A TMP36 turns temperature into a changing voltage that a program can measure.

Its output rises by 10 millivolts for every degree Celsius, with 0.5 V representing 0 °C.

The Pico reads that voltage directly; the Raspberry Pi uses an MCP3008 analogue-to-digital converter.

Uses

Refrigerator cooling

DETECT Measure fridge temperature → **DECIDE** Is it too warm? → **DO** Increase the cooling

Soup drama meter

DETECT Measure bowl surface → **DECIDE** Which drama level fits? → **DO** Move the drama pointer

Flat face	+Vs	Vout	GND
Use it to identify the pins	Left pin	Centre pin	Right pin

Raspberry Pi Foundation

Code

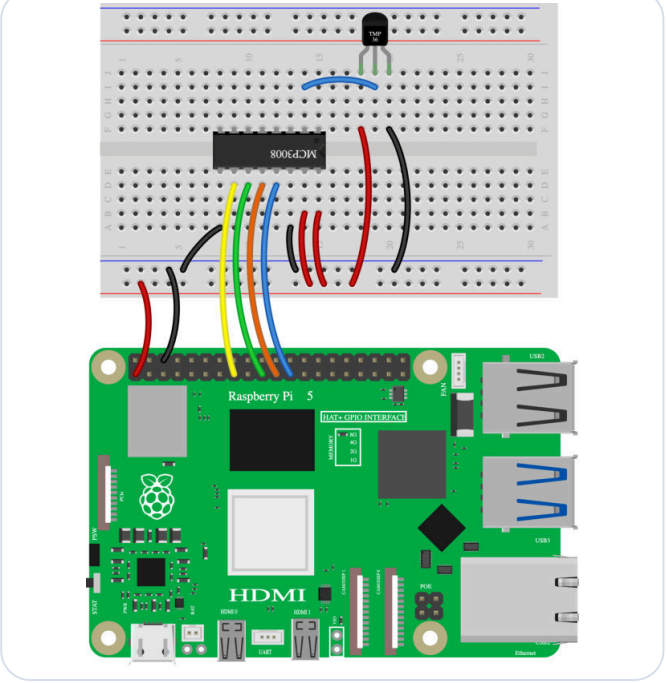
Raspberry Pi

```
from gpiozero import MCP3008

sensor = MCP3008(channel=1, select_pin=22)

temperature = (sensor.voltage - 0.5) * 100
```

Read the TMP36 with its flat face towards you; the pins are +Vs, Vout, GND from left to right.

Wiring

RASPBERRY PI 5



TMP36 **Vout** → MCP3008 **CH1**. CLK / DIN / DOUT / CS → **GPIO 11 / 10 / 9 / 22** (pins 23 / 19 / 21 / 15). **3V3** (pin 1), **GND** (pin 6).

Temperature probe

DETECT | RASPBERRY PI | DS18B20 | DIGITAL

What does it do?

A **DS18B20** measures temperature and sends a digital reading in degrees Celsius.

This probe has a sealed metal tip on a cable with three wires. Check the supplier's pinout; wire colours can vary.

Only the sealed probe goes in water. Keep the board and all wire connections dry.

Uses

Aquarium temperature alarm

DETECT Measure tank temperature → **DECIDE** Is it too cold? → **DO** Show a warning

Penguin bath alarm

DETECT Measure bath temperature → **DECIDE** Is it too warm? → **DO** Light a warning LED

VDD	DQ	GND	4.7 kΩ
Power from 3V3	Digital data	Ground	Resistor from DQ to 3V3

Raspberry Pi Foundation

Code

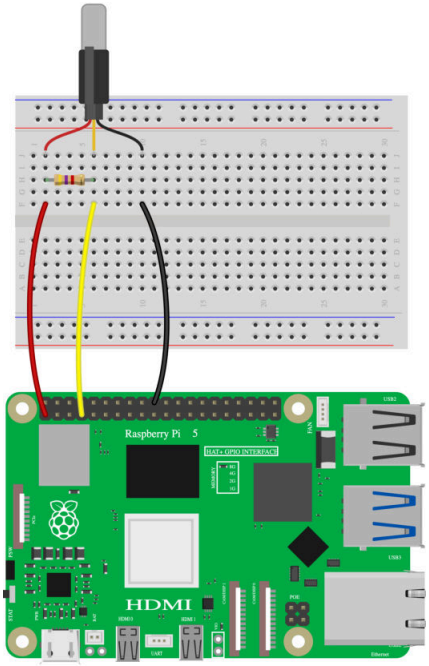
Raspberry Pi

```
from w1thermsensor import W1ThermSensor

sensor = W1ThermSensor()
temperature = sensor.get_temperature()
print(temperature)
```

Setup: enable 1-Wire in `raspi-config`, reboot, then run `sudo apt install python3-w1thermsensor`.

Wiring

RASPBERRY PI 5



VDD → 3V3 (pin 1); **DQ** → GPIO4 (pin 7); **GND** → GND (pin 20). **4.7 kΩ** between DQ and 3V3.

Tilt switch

DETECT | RASPBERRY PI | INPUT

What does it do?

A tilt switch changes state when it tips past a particular angle.

While it stands upright, a tiny metal ball rests on two contacts and completes the circuit. Tip it over and the ball rolls away, breaking the circuit.

It can detect that an object has been picked up, tipped over or opened.

Uses

Iron safety shut-off

DETECT: Iron tips sideways → DECIDE: Has the delay elapsed? → DO: Switch the heater off

Grumpy pencil case

DETECT: Case is tipped → DECIDE: Has it stayed tipped? → DO: Make a grumpy noise

Metal can
Contains the rolling ball

Legs
No polarity

Angle
Changes the contact

Raspberry Pi Foundation

Code

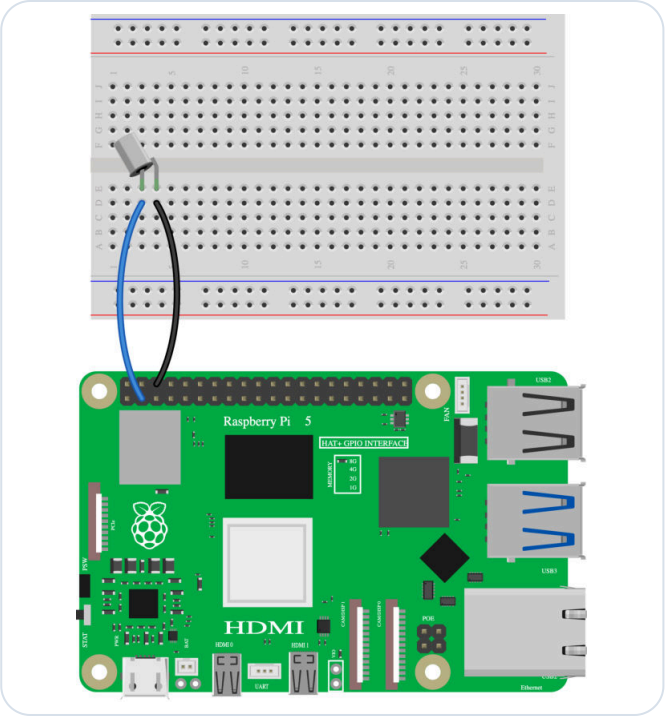
Raspberry Pi

```
from gpiozero import Button

tilt = Button(2)

tilted = not tilt.is_pressed
```

> The reading may flicker while the ball rolls. Check it more than once before treating it as a definite tilt.

Wiring

RASPBERRY PI 5



GPIO 2 (pin 3) → tilt switch → **GND** (pin 6)

Toggle switch

DETECT | RASPBERRY PI | INPUT

What does it do?

A toggle switch stays in the position you move it to, unlike a push button.

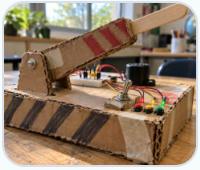
Its centre COM leg connects to one outside leg in each position.

It is useful when a project needs a lasting on/off choice or two modes.

Uses

Auto/manual selector

DETECT Read selector position → **DECIDE** Is automatic mode selected? → **DO** Run the selected mode



Serious business lever

DETECT Lever and check buttons → **DECIDE** Are all checks complete? → **DO** Start unnecessary sirens

Lever
Stays where it is moved

COM
Centre pin

L1 and L2
The two choices

Raspberry Pi Foundation

Code

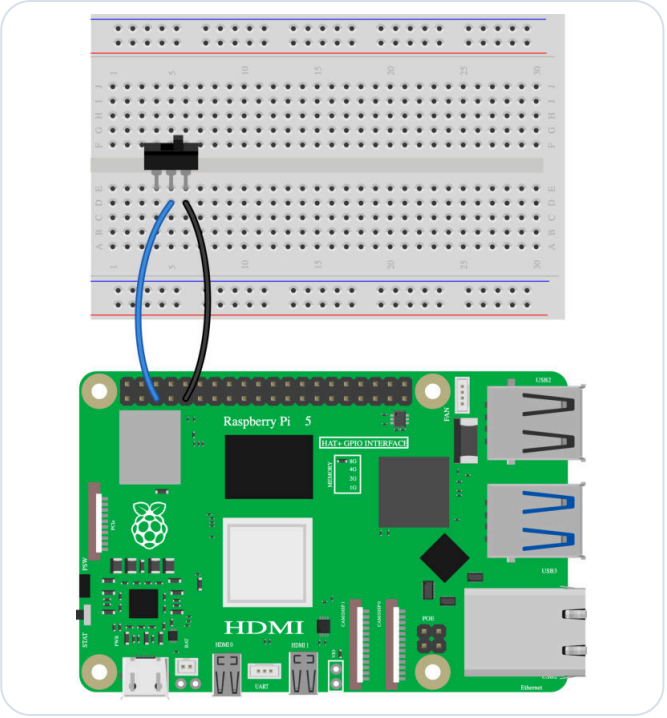
Raspberry Pi

```
from gpiozero import Button

mode = Button(3)

mode_two = mode.is_pressed
```

Connect COM and one outside pin. The unused outside pin can be left unconnected.

Wiring

RASPBERRY PI 5



GPIO 3 (pin 5) → COM; right-hand outside pin → **GND** (pin 9)

Pi Camera

DETECT | RASPBERRY PI | IMAGES | VIDEO

What does it do?

A camera turns light into digital images. Connect a Raspberry Pi Camera Module to a Raspberry Pi computer with a flat ribbon cable.

Use Python to take a photograph or record a video. Your program can also examine images and decide what to do next.

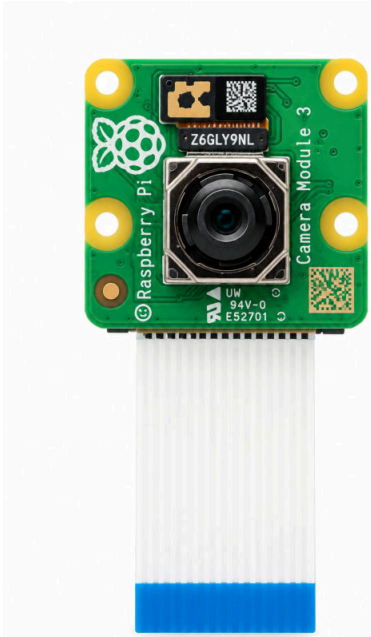
Uses

Video doorbell

DETECT Camera sees movement. → **DECIDE** Is a person there? → **DO** Save a video clip.

Monster photo booth

DETECT Button starts countdown. → **DECIDE** Has time run out? → **DO** Take a silly photo.



Lens
Focuses light

Sensor
Records the image

Ribbon
Connects to the Pi

Code

Take photo

```
from picamera2 import Picamera2

camera = Picamera2()
camera.start_and_capture_file(
    "photo.jpg", delay=2,
    show_preview=False)
camera.close()
```

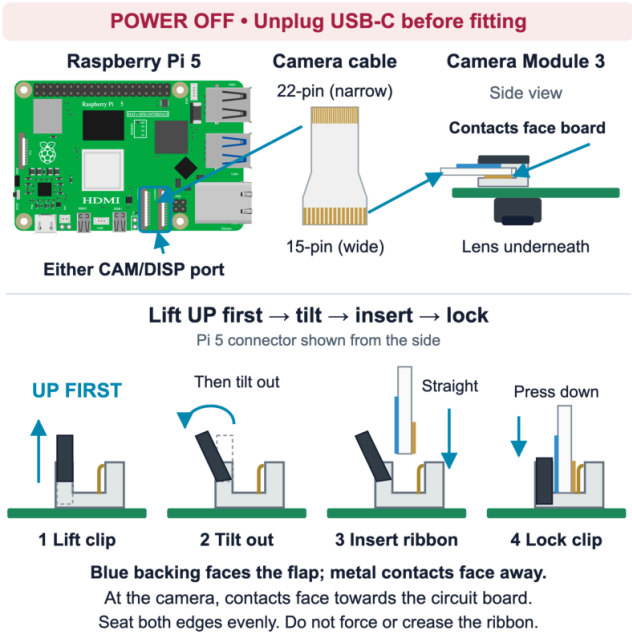
Take video

```
from picamera2 import Picamera2

camera = Picamera2()
camera.start_and_record_video(
    "video.mp4", duration=5)
camera.close()
```

Run each example separately on **Raspberry Pi OS** with **Picamera2** installed. Files save in the current folder. The video is five seconds long, without sound.

Wiring

RASPBERRY PI 5 + CAMERA MODULE 3



Use a **Standard–Mini camera cable** (22-pin to 15-pin). The wide cable supplied with the camera fits a Pi 4, not a Pi 5.

LED

DO | RASPBERRY PI | LIGHT

What does it do?

An LED is a tiny light that your code can switch on, switch off, or blink.

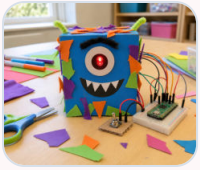
LED stands for light-emitting diode. It glows when electricity flows the right way through it, so the long leg and the short leg matter.

Your projects can use LEDs to show a warning, celebrate something, or tell people what is happening.

Uses

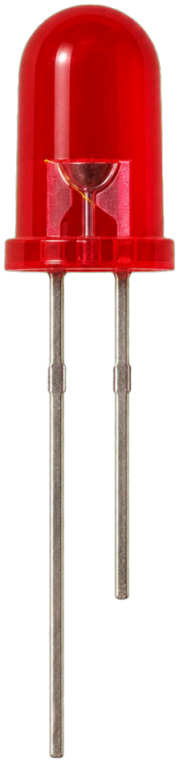
Car brake light

DETECT Brake control signal → **DECIDE** Is braking requested? → **DO** Light the red LEDs



One-eyed monster box

DETECT Box lid is opened → **DECIDE** Is guard mode armed? → **DO** Glow the monster eye



Anode
Long leg, to GPIO

Cathode
Short leg, to GND

Lens
Glows here

Raspberry Pi Foundation

Code

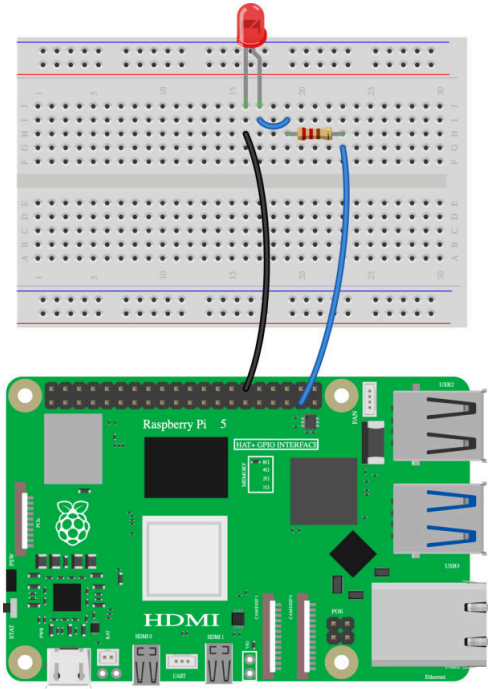
Raspberry Pi

```
from gpiozero import LED

led = LED(26)

led.on()
```

Also `led.off()`, `led.toggle()` and `led.blink()`. Give `blink()` an `on_time` and an `off_time` to set the flash speed.

Wiring

RASPBERRY PI 5



GPIO 26 (pin 37) → 220 Ω → LED → **GND** (pin 30)

RGB LED

DO | RASPBERRY PI | OUTPUT | LIGHT

What does it do?

An RGB LED puts red, green and blue lights inside one clear package.

Varying the brightness of the three colours mixes millions of possible colours.

A common-cathode LED has one shared negative leg and needs one resistor for each colour.

Uses

Wi-Fi router status

DETECT Read connection status → **DECIDE** Which status colour is needed? → **DO** Mix the LED colour

Mood crystal

DETECT Read the mood dial → **DECIDE** What is the crystal's mood? → **DO** Glow that colour

Clear lens
Contains three LEDs

Long leg
Common cathode

R, G and B legs
One 220 ohm resistor each

Raspberry Pi Foundation

Code

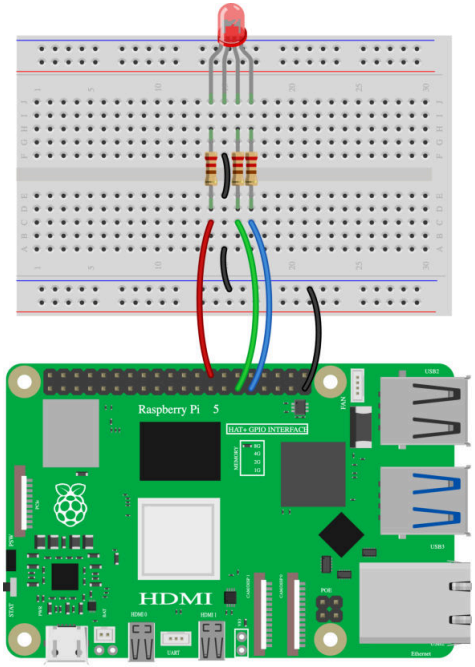
Raspberry Pi

```
from gpiozero import RGBLED

light = RGBLED(red=7, green=5, blue=6)

light.color = (1, 0, 1)
```

Colour values run from 0 to 1. This example makes magenta.

Wiring

RASPBERRY PI 5



R → **GPIO7** (pin 26), G → **GPIO5** (pin 29), B → **GPIO6** (pin 31); common → **GND** (pin 39)

NeoPixels

DO | RASPBERRY PI | OUTPUT | LIGHT

What does it do?

A NeoPixel is a full-colour LED with a tiny controller built into it.

One data wire carries colour instructions, and each pixel passes the remaining instructions onwards.

The 3.3 V-compatible JST breakout shown here can connect directly to a Pi or Pico.

Uses

Gaming PC lighting

DETECT Read lighting selection → **DECIDE** Which colour should each pixel be? → **DO** Light individual pixels

Disco shoelace meter

DETECT Read shoe movement → **DECIDE** Which light pattern should run? → **DO** Animate the pixels

Pixel RGB LED and controller

IN Data from the board

OUT Data to another pixel

VDD and GND 3.3 V power

Raspberry Pi Foundation

Code

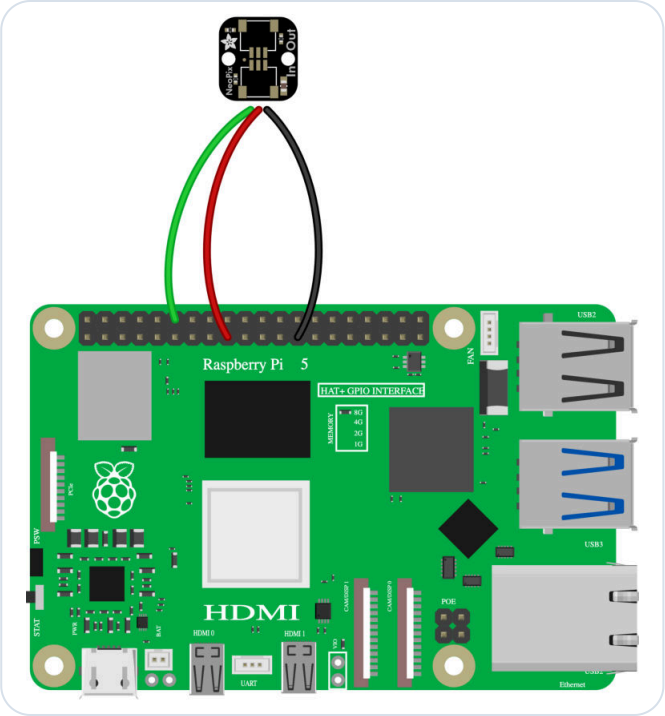
Raspberry Pi

```
import board
import neopixel

pixels = neopixel.NeoPixel(board.D18, 1)

pixels[0] = (255, 0, 80)
```

This wiring is for the pictured 3.3 V-compatible breakout. Generic 5 V strips usually need a separate supply and a logic-level shifter.

Wiring

RASPBERRY PI 5



GPIO18 (pin 12) → IN, **3V3** (pin 17) → VDD, **GND** (pin 25) → GND

Active buzzer

DO | RASPBERRY PI | SOUND

What does it do?

An active buzzer makes one built-in tone. Switch it on and off for warning beeps or rhythms; its pitch stays the same.

Recognise it: check the packet for **active** or **built-in oscillator**. Shape, stickers and + marks do not prove the type.

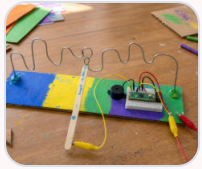
Use a labelled **3.3 V GPIO-compatible** buzzer.

Uses

Smoke alarm

One warning tone is enough; no changing notes needed.

DETECT Measure smoke level → **DECIDE** Is smoke above the limit? → **DO** Repeat one warning tone



Buzz wire game

Choose active: one fixed buzz clearly signals failure.

DETECT Loop touches wire → **DECIDE** Is this the third strike? → **DO** Buzz at one pitch

+ leg
Marked +, to GPIO

– leg
Other leg, to GND

Vent
Sound comes out

Raspberry Pi Foundation

Code

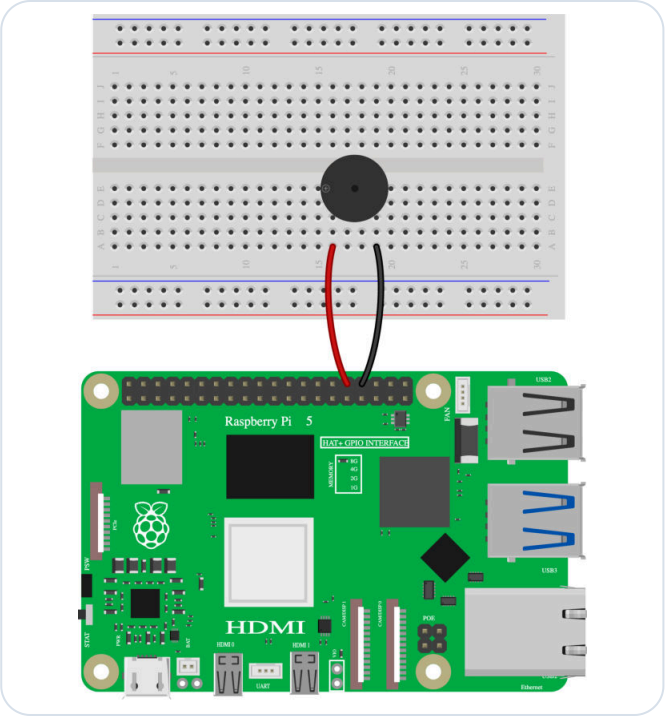
Raspberry Pi

```
from gpiozero import Buzzer

buzzer = Buzzer(12)

buzzer.on()
```

Also `buzzer.off()`, `buzzer.toggle()` and `buzzer.beep()`. Give `beep()` an `on_time` and an `off_time` to set the rhythm. Faster beeps do not mean a higher pitch.

Wiring

RASPBERRY PI 5



GPIO 12 (pin 32) → **+** buzzer **−** → **GND** (pin 34)

Passive buzzer

DO | RASPBERRY PI | SOUND | PWM

What does it do?

A passive buzzer needs fast pulses (PWM) to make sound. Changing their frequency changes its pitch: low or high notes.

Recognise it: check for **passive**, **external drive** or **no oscillator** on the packet. Both types can look identical; + only marks polarity.

Use a small **3.3 V GPIO-compatible piezo** sounder.

Uses

Washing-machine chime

PWM plays tunes; active buzzers have only one pitch.

DETECT Read cycle status → **DECIDE** Has the wash finished? → **DO** Play a short tune

Robot voice box

Change pitch for happy chirps or grumpy growls.

DETECT Read mood button → **DECIDE** Is happy pressed? → **DO** Chirp high or low

+ leg
Marked +, to GPIO

– leg
Other leg, to GND

Vent
Sound comes out

Raspberry Pi Foundation

Code

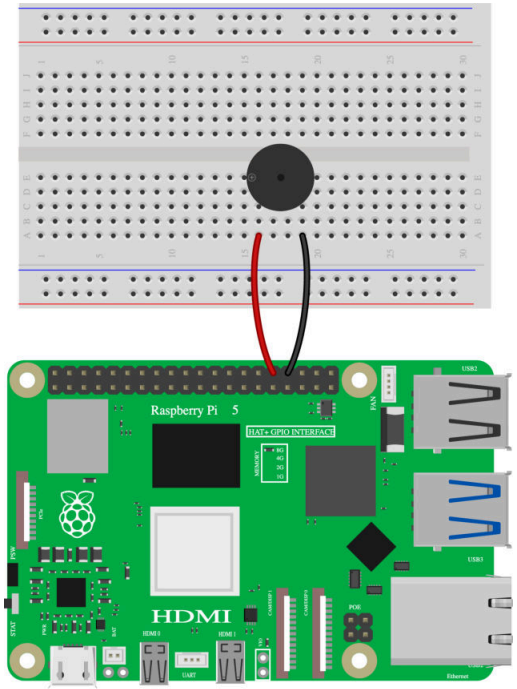
Raspberry Pi

```
from gpiozero import TonalBuzzer

buzzer = TonalBuzzer(12)

buzzer.play("A5")
```

Try `buzzer.play("A4")`; silence with `buzzer.stop()`. Higher frequency means higher pitch. Beep speed is separate; both types can make rhythms.

Wiring

RASPBERRY PI 5



GPIO 12 (pin 32) → **+** buzzer **−** → **GND** (pin 34)

Speaker

DO | RASPBERRY PI | OUTPUT | SOUND

What does it do?

A small passive speaker turns a changing electrical signal into sound.

Current through its coil moves a cone backwards and forwards, pushing the air.

The 220 ohm resistor shown here protects the GPIO; it also keeps this simple circuit fairly quiet.

Uses

Door chime

DETECT Button and chime setting → **DECIDE** Which chime should play? → **DO** Play the notes

Tiny fanfare machine

DETECT Count achievement-button presses → **DECIDE** Has the target been reached? → **DO** Play triumphant notes

Cone
Pushes the air

Coil and magnet
Create movement

Terminals
Connect through 220 ohms

Raspberry Pi Foundation

Code

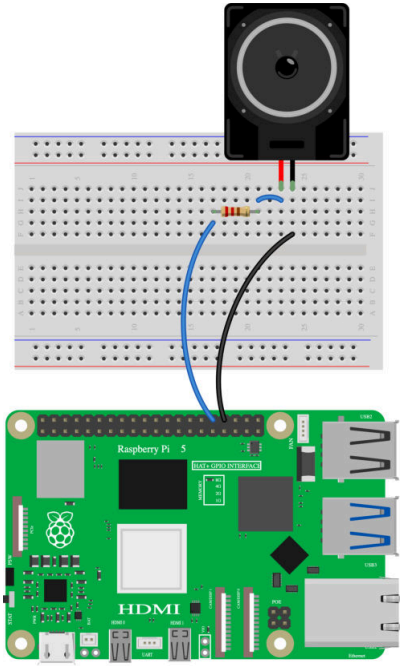
Raspberry Pi

```
from gpiozero import TonalBuzzer

speaker = TonalBuzzer(12)

speaker.play("E4")
```

For a louder speaker, use a transistor or amplifier rather than removing the protective resistor.

Wiring

RASPBERRY PI 5



GPIO12 (pin 32) → 220 ohms → speaker red (+) lead; black lead → **GND** (pin 34)

Motor driver

DO | RASPBERRY PI | OUTPUT | MOVEMENT

What does it do?

A motor driver lets a low-power GPIO control a motor that needs more current.

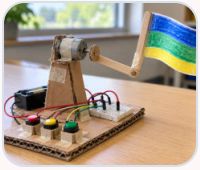
The L293D contains H-bridges that can send current through a motor in either direction.

Two input pins choose forwards, backwards or stop, while the chip handles the motor current.

Uses

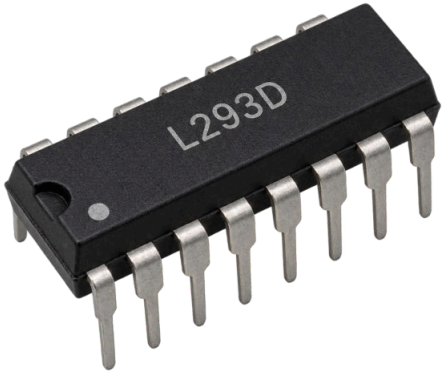
Electric car window

DETECT Read window switch → **DECIDE** Can it move safely? → **DO** Drive requested direction



Tiny flag waver

DETECT Count score-button presses → **DECIDE** Has the target been reached? → **DO** Wave the flag



Notch
Identifies pin 1

IN1 and IN2
Direction controls

OUT1 and OUT2
Connect to the motor

Enable
Held high in this circuit

Code

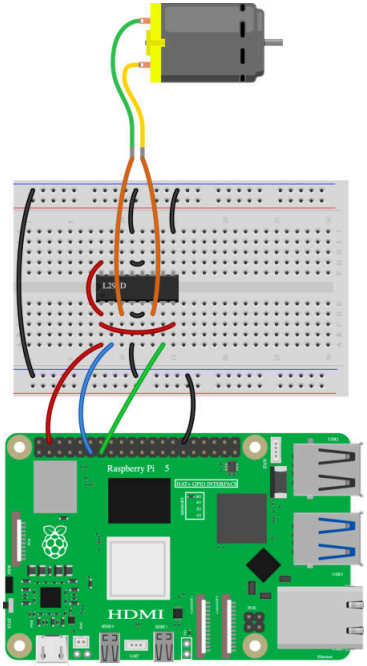
Raspberry Pi

```
from gpiozero import Motor

motor = Motor(forward=17, backward=27)

motor.forward()
```

This card powers one small 5 V motor from the board's 5 V rail. Use a separate motor supply for larger motors, with the grounds joined.

Wiring

RASPBERRY PI 5



GPIO17 (pin 11) → IN1, **GPIO27** (pin 13) → IN2; OUT1 and OUT2 → motor; **5V** (pin 4), **GND** (pin 30)

Servo motor

DO | RASPBERRY PI | OUTPUT | MOVEMENT

What does it do?

A positional servo moves its arm, called a horn, to a chosen position and holds it there.

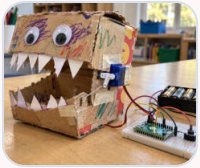
Short pulses on its signal wire set the position. The motor and position control are built into the case.

Use it for steering, pointing or opening a flap. A continuous-rotation servo controls speed instead; it cannot use this example to choose an angle.

Uses

Radio-controlled steering

DETECT Read steering command → **DECIDE** Which way should it turn? → **DO** Set the steering position



Cardboard chomping monster

DETECT Read feeding button → **DECIDE** Is the monster hungry? → **DO** Open its cardboard jaw

Horn	Signal	Power	Ground
Attach a light moving part	Usually orange, yellow or white	Usually red; about 5 V	Brown or black; join grounds

Raspberry Pi Foundation

Code

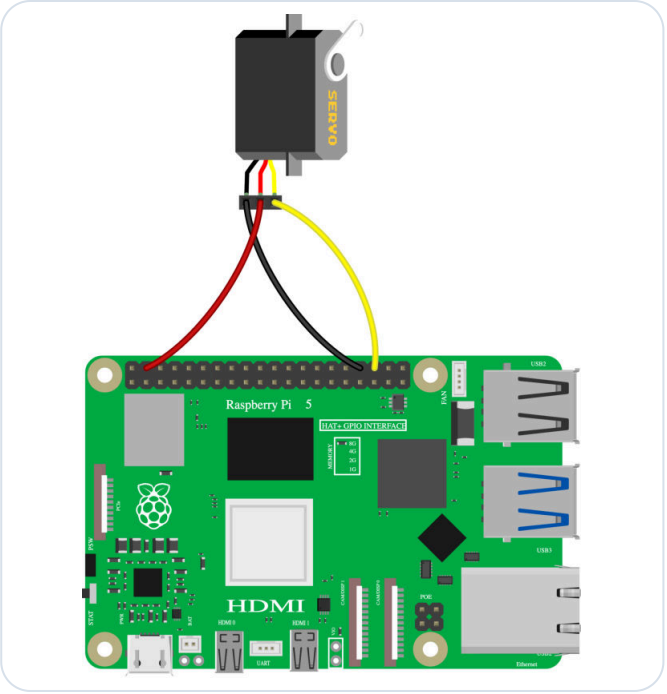
Raspberry Pi

```
from gpiozero import Servo
from time import sleep

servo = Servo(16)
servo.mid()
sleep(2)
```

Test unloaded: the horn centres. Try min() or max() instead of mid(). Travel varies; stop if it binds or buzzes. Check lead order.

Wiring

RASPBERRY PI 5



GPIO16 (pin 36) → signal; **5V** (pin 4) → servo red; **GND** (pin 34) → servo −. Small **SG90-size** servo only: the Pi powers it.

Stepper motor

DO | RASPBERRY PI | OUTPUT | MOVEMENT

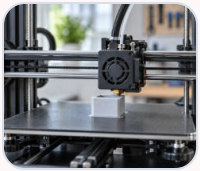
What does it do?

A stepper motor turns in small, repeatable steps instead of spinning freely.

Energising its coils in sequence moves the shaft one step at a time and holds its position.

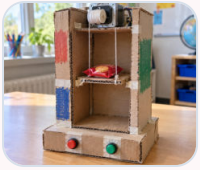
The ULN2003 board supplies the coil current for the common 28BYJ-48 motor shown here.

Uses



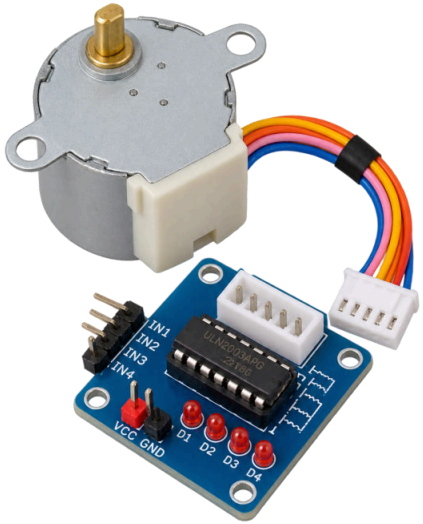
3D printer positioning

DETECT Read next print position → **DECIDE** How many steps are needed? → **DO** Move the print head



Snack lift

DETECT Snack request arrives → **DECIDE** Which floor needs it? → **DO** Step lift into place



Output shaft Moves in measured steps

Five-wire plug Connects to the driver

IN1 to IN4 Coil sequence

5V and GND Driver power

Code

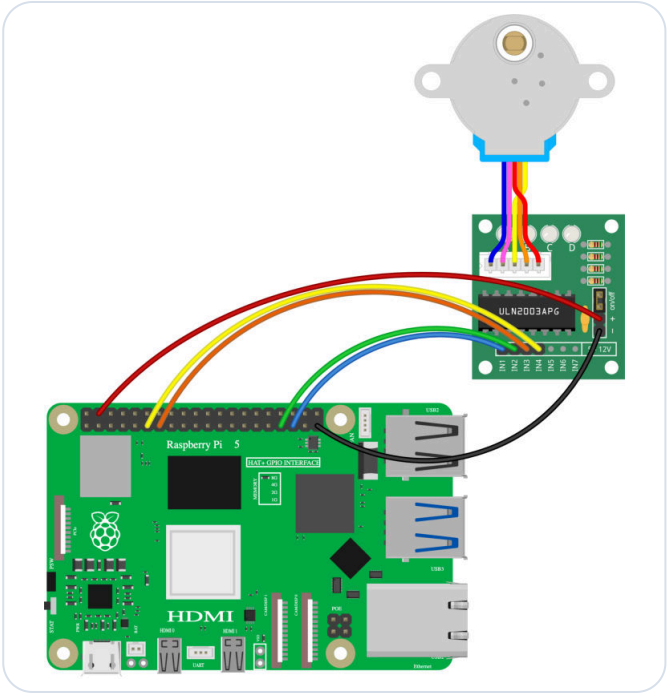
Raspberry Pi

```
from gpiozero import OutputDevice
from time import sleep

coils = [OutputDevice(pin) for pin in (19, 13, 27, 17)]
sequence = ((1, 0, 0, 0), (0, 1, 0, 0),
            (0, 0, 1, 0), (0, 0, 0, 1))

for pattern in sequence * 512:
    for coil, value in zip(coils, pattern):
        coil.value = value
    sleep(0.002)
```

`sequence * 512` runs about one full turn. Reverse the sequence to turn the other way.

Wiring

RASPBERRY PI 5



IN1-IN4 → GPIO **19, 13, 27, 17** (pins 35, 33, 13, 11); driver → **5V** (pin 4) and **GND** (pin 39)

Robot / rover

DO | RASPBERRY PI | OUTPUT | MOVEMENT

What does it do?

A two-wheel rover uses a separate geared motor for each wheel. A caster or smooth skid supports the other end.

Drive both wheels forwards to move; change their speeds or directions to turn. A motor driver handles their power.

The Robot class joins the two motors into one object. This example uses a breadboard-mounted L293D chip and a separate motor battery pack.

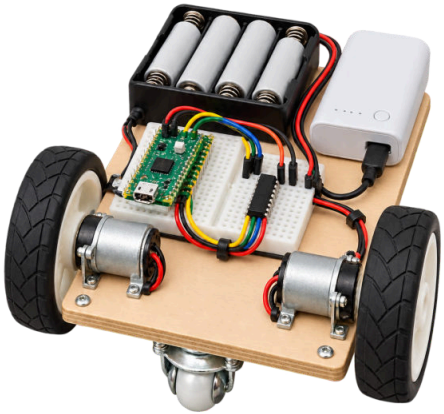
Uses

Robot vacuum cleaner

DETECT Sense a nearby obstacle → **DECIDE** Is the path clear? → **DO** Drive on or turn away

Sock delivery turtle

DETECT Read delivery button → **DECIDE** Is a delivery requested? → **DO** Drive the sock parcel



Drive wheels
One geared motor per side

Motor driver
L293D, a 16-pin breadboard chip

Caster or skid
Supports the chassis

Power
Separate motor and USB supplies

Raspberry Pi Foundation

Code

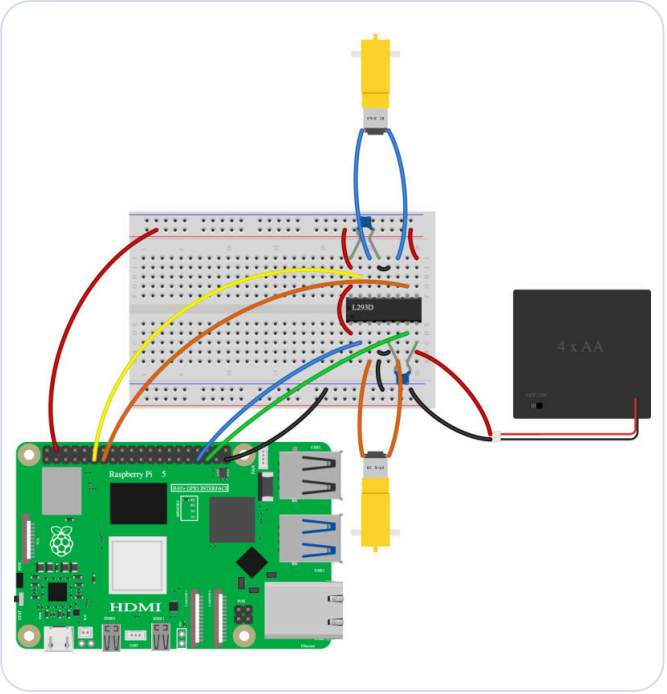
Raspberry Pi

```
from gpiozero import Motor, Robot
from time import sleep

robot = Robot(left=Motor(13, 19),
              right=Motor(17, 27))
try:
    robot.forward()
    sleep(1)
finally:
    robot.stop()
```

Test with wheels lifted. Reverse a motor by swapping its leads with power off. Try left(), right(), backward() and stop().

Wiring

RASPBERRY PI 5



GPIO13/19 (Pi pins 33/35) → L293D pins 2/7; **GPIO17/27** (Pi pins 11/13) → 15/10. **5 V** (Pi pin 4) → 1, 9, 16. **GND** (Pi pin 39) → 4, 5, 12, 13.